

OBJECTIVE TYPE QUESTIONS

In this appendix, objective type questions pertaining to Chapters 2 to 10 are given. In all the chapters, the questions are so framed that the answer for each may be a number, a clause or a sentence. In case four alternatives are provided, there is one correct answer unless otherwise stated in the question.

POWER SEMICONDUCTOR DIODES AND TRANSISTORS

1. For a diode, reverse recovery time is defined as the time between the instant diode current becomes zero and the instant reverse recovery current decays to
 - (a) zero
 - (b) 10% of reverse peak current I_{RM}
 - (c) 25% of I_{RM}
 - (d) 15% of I_{RM}
2. In a diode, the cut-in voltage and forward-voltage drop are respectively
 - (a) 0.7 V, 0.7 V
 - (b) 0.7 V, 1 V
 - (c) 0.7 V, 0.6 V
 - (d) 1 V, 0.7 V
3. The softness factor for soft-recovery and fast-recovery diodes are respectively
 - (a) 1, > 1
 - (b) < 1, 1
 - (c) 1, 1
 - (d) 1, < 1
4. Reverse recovery current in a diode depends upon
 - (a) forward field current
 - (b) storage charge
 - (c) temperature
 - (d) PIV
5. In a BJT,
 - (a) $\beta = \frac{\alpha}{\alpha + 1}$
 - (b) $\beta = \frac{\alpha}{\alpha - 1}$
 - (c) $\alpha = \frac{\beta}{\beta - 1}$
 - (d) $\alpha = \frac{\beta + 1}{\beta}$
6. A power MOSFET has three terminals called
 - (a) collector, emitter and base
 - (b) drain, source and base
 - (c) drain, source and gate
 - (d) collector, emitter and gate
7. As compared to power MOSFET, a BJT has
 - (a) lower switching losses but higher conduction loss
 - (b) higher switching losses and higher conduction loss
 - (c) higher switching losses but lower conduction loss
 - (d) lower switching losses and lower conduction loss
8. Choose the correct statement :
 - (a) MOSFET has positive temperature coefficient (TC) whereas BJT has negative TC
 - (b) Both MOSFET and BJT have positive TC
 - (c) Both MOSFET and BJT have negative TC
 - (d) MOSFET has negative TC whereas BJT has positive TC
9. Choose the correct statement :
 - (a) Both MOSFET and BJT are voltage controlled devices (CDs)
 - (b) Both MOSFET and BJT are current CDs

Appendix

- (c) MOSFET is a voltage CD whereas BJT is a current CD
 (d) MOSFET is a current CD whereas BJT is a voltage CD
10. Secondary breakdown occurs in
 (a) MOSFET but not in BJT (b) both MOSFET and BJT
 (c) BJT but not in MOSFET (d) none of these
11. At present, the state of the art devices are available as under :
- | MOSFET | BJT |
|-------------------|----------------|
| (a) 1200 V, 800 A | 500 V, 140 A |
| (b) 500 V, 140 A | 1200 V, 800 A |
| (c) 800 V, 1000 A | 1000 V, 1200 A |
| (d) 200 V, 140 A | 1500 V, 800 A |
12. An IGBT has three terminals called
 (a) collector, emitter and base (b) drain, source and base
 (c) drain, source and gate (d) collector, emitter and gate
13. An MCT has three terminals called
 (a) anode, cathode and gate (b) collector, emitter and gate
 (c) drain, source and base (d) drain, source and gate
14. For the switching waveform shown Fig. A-1 in for a power transistor, the peak instantaneous power loss in watts is
 (a) 250 (b) 500
 (c) 166.67 (d) 333.33
15. For the switching waveform shown in Fig. A-1 for a power transistor, the average value of switch-on power loss at a switching frequency of 10 kHz is
 (a) 1 W (b) 2 W
 (c) 3 W (d) 4 W
16. Match the devices on the left hand side with the circuit symbols on right hand side.

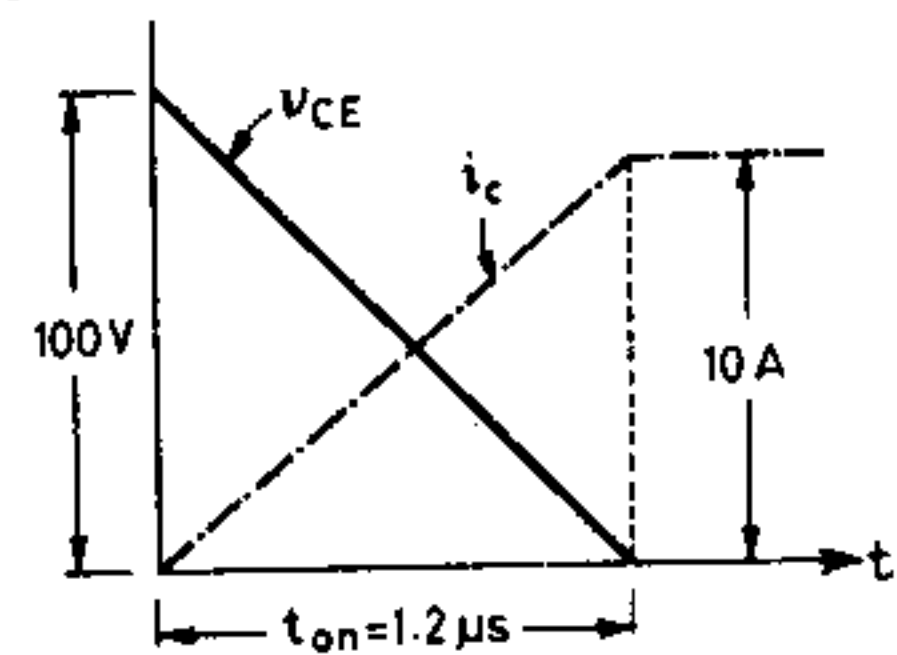


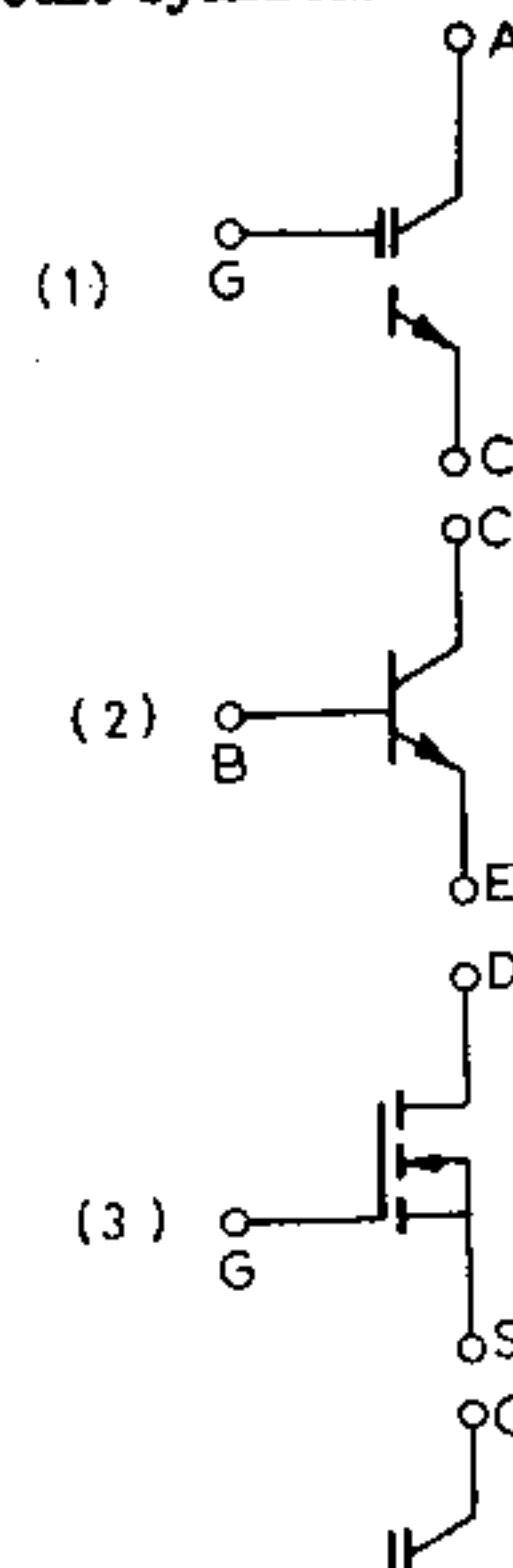
Fig. A-1

(A) BJT

(B) MOSFET

(C) IGBT

(D) MCT



ANSWERS

- | | | | | | |
|---------|---------|---------|---------------------------|---------|---------|
| 1. (c) | 2. (c) | 3. (d) | 4. (a) | 5. (b) | 6. (c) |
| 7. (c) | 8. (a) | 9. (c) | 10. (c) | 11. (b) | 12. (d) |
| 13. (a) | 14. (a) | 15. (b) | 16. A-2 ; B-3 ; C-4 ; D-1 | | |

DIODE CIRCUITS AND RECTIFIERS

1. In Fig. A.2, capacitor C is charged to $V_0 = 50$ V with upper plate positive. Switch S is closed at $t = 0$. Current through the circuit at $t = 0$ and final voltage across C are respectively,

- | | |
|-----------------|-----------------|
| (a) 15 A, 200 V | (b) 20 A, 200 V |
| (c) 25 A, 250 V | (d) 15 A, 150 V |

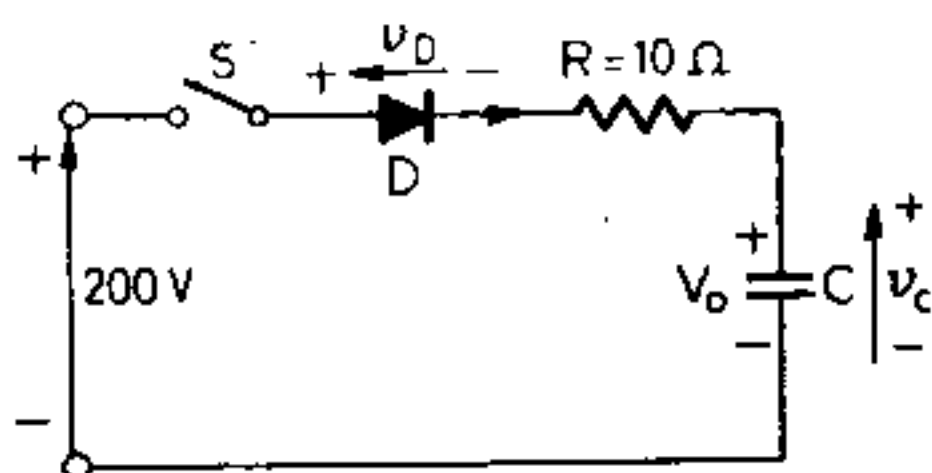


Fig. A-2

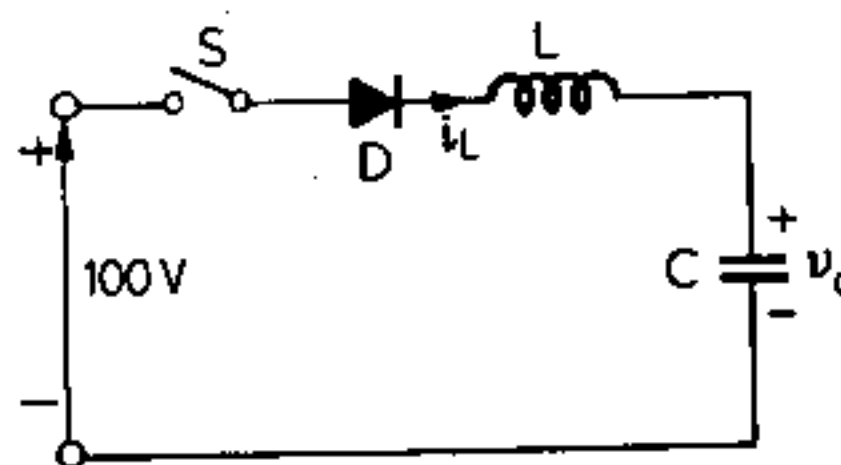


Fig. A-3

2. In Fig. A.2, suppose capacitor C is charged to 50 V with lower plate positive. Switch S is closed at $t = 0$. Current through the diode at $t = 0$ and final voltage across C are respectively

- | | |
|-----------------|-----------------|
| (a) 25 A, 250 V | (b) 25 A, 200 V |
| (c) 20 A, 200 V | (d) 15 A, 150 V |

3. In the circuit of Fig. A.3, switch S is closed at $t = 0$ with $i_L(0) = 0$ and $v_c(0) = 0$. In steady-state, v_c equals

- | | |
|-----------|------------|
| (a) 200 V | (b) 100 V |
| (c) zero | (d) -100 V |

4. In Fig. A.4 ; V_1 , V_2 and V_3 are zero centre PMMC voltmeters. The circuit is initially relaxed. Switch S is closed at $t = 0$. In steady state, readings of voltmeters V_1 , V_2 and V_3 are respectively

- | | |
|--------------------------|---------------------|
| (a) 100 V, 100 V, -100 V | (b) 100 V, 0, 200 V |
| (c) -100 V, 0, 200 V | (d) 100 V, 0, 100 V |

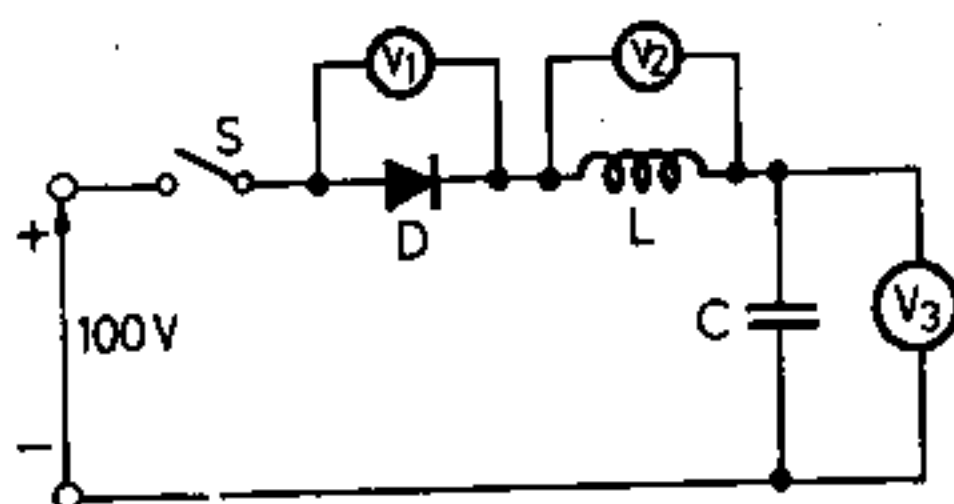


Fig. A-4

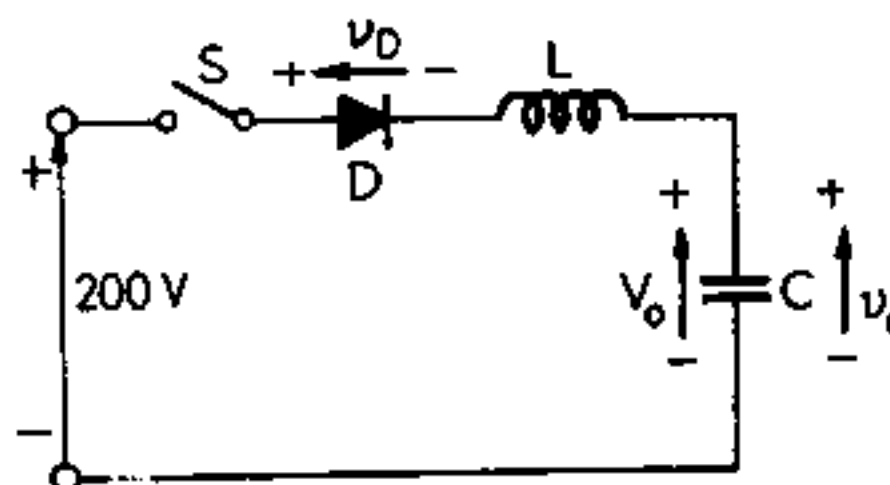


Fig. A-5

5. In Fig. A.5, initial voltage across capacitor is $V_0 = 50$ V with the polarity as shown. Switch S is closed at $t = 0$. In steady state, v_c and v_D are respectively given by

Appendix

- (a) 400 V, -200 V (b) 350 V, -150 V
 (c) 200 V, 0 (d) 450 V, -250 V
6. In Fig. A.5, if $C = 8 \mu\text{F}$ and $L = 0.2 \text{ mH}$, the peak current handled by diode is
 (a) 40 A (b) 50 A
 (c) 10 A (d) 30 A

7. In Fig. A.6, initial voltage across capacitor is $V_0 = 50 \text{ V}$ with the polarity as shown. Switch S is closed at $t = 0$. In steady-state, v_C and v_D are respectively given by

- (a) 400 V, -200 V (b) 350, -150 V
 (c) 200 V, 0 (d) 450 V, -250 V

8. In Fig. A.6, if $C = 8 \mu\text{F}$ and $L = 0.2 \text{ mH}$, the peak current through diode is given by

- (a) 40 A (b) 50 A
 (c) 10 A (d) 30 A

9. In Fig. A.7, capacitor C is initially charged with voltage V_0 with upper plate positive. Switch S is closed at $t = 0$. At $t = 0+$, v_C and i are given by

- (a) $0, \frac{V_0}{R}$ (b) $-V_0, \frac{V_0}{R}$
 (c) $-V_0, -\frac{V_0}{R}$ (d) $V_0, \frac{V_0}{R}$

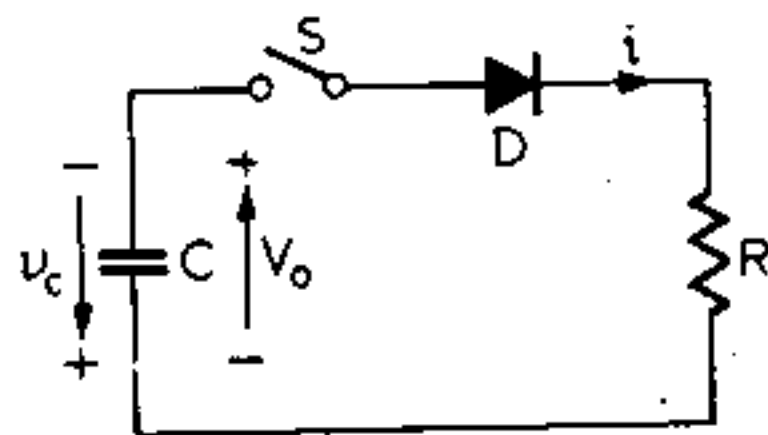


Fig. A-7

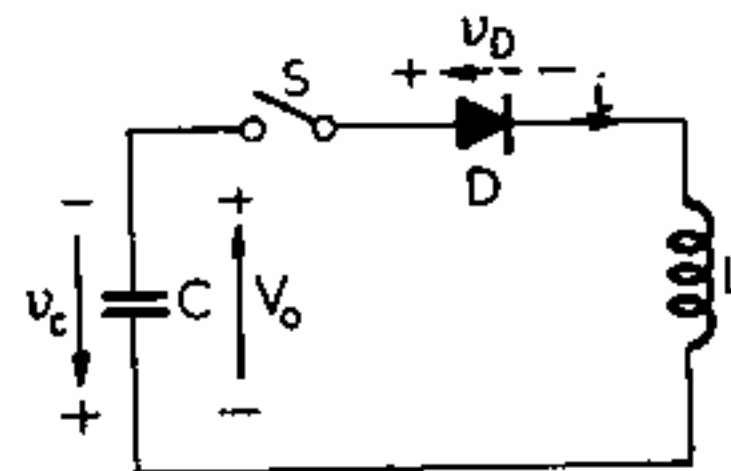


Fig. A-8

10. In Fig. A.8, capacitor C is initially charged with voltage V_0 . Switch S is closed at $t = 0$. In steady state, v_C and v_D are respectively given by

- (a) $V_0, -V_0$ (b) $0, -V_0$
 (c) $-V_0, 0$ (d) V_0, V_0

11. Each diode in Fig. A.9 can be described by a cut-in voltage and zero resistance. If the cut-in voltage of diode D1 is 0.2 V and of diode D2 is 0.6 V, the magnitude of current I_1 through D1 is mA and magnitude of current through D2 is mA.

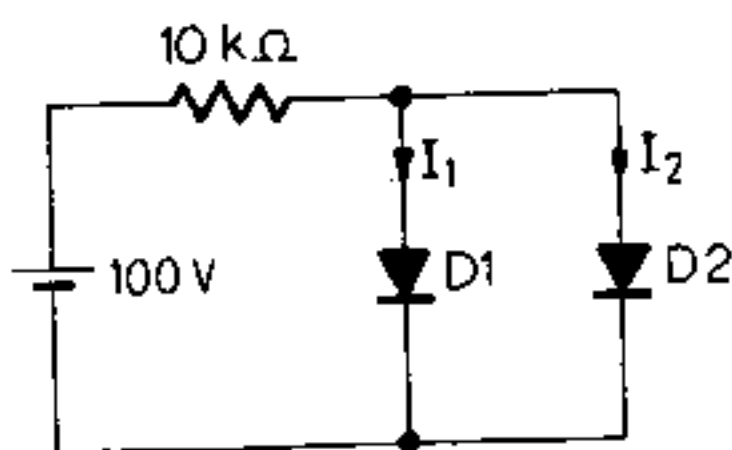


Fig. A-9

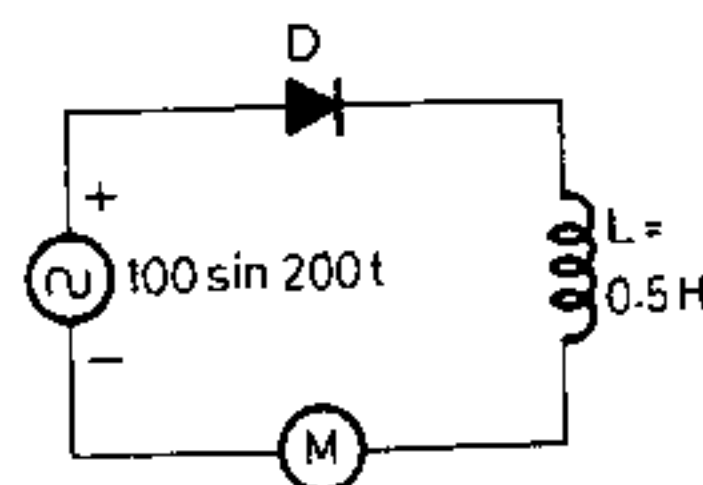


Fig. A-10

12. In Fig. A.10, ideal PMMC ammeter M will read

- (a) zero
(c) 0.707 A
- (b) 1.414 A
(d) 1 A
13. In Fig. A.10, if ammeter M is an ideal MI ammeter, then it will read
(a) 0.707 A
(b) 1.414 A
(c) 1.225 A
(d) 1 A
14. In Fig. A.11, ideal moving iron voltmeters $M1$ and $M2$ will respectively read
(a) 141.4 V, 141.4 V
(b) 0, 141.4 V
(c) 0, 200 V
(d) 141.4 V, 0

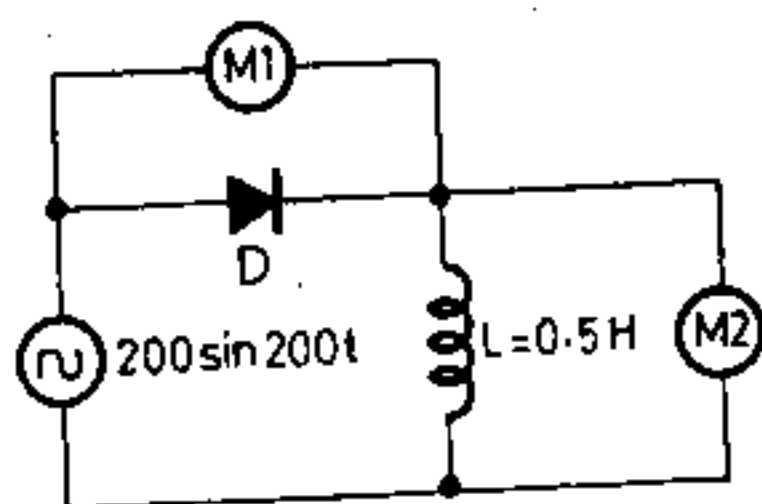


Fig. A-11

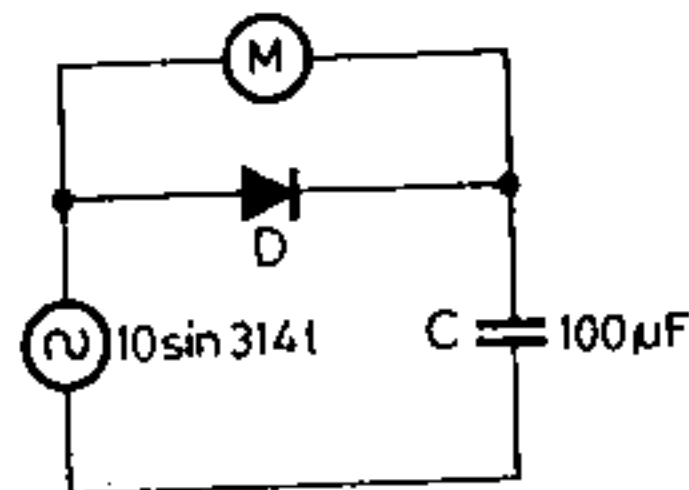


Fig. A-12

15. In Fig. A.12, an ideal moving iron voltmeter M will read
(a) 7.07 V
(b) 12.25 V
(c) 14.14 V
(d) 20.0 V
16. In Fig. A.13, zero-centre and ideal PMMC voltmeters $M1$ and $M2$ will read
(a) -10 V, 10 V
(b) 0, 10 V
(c) -10 V, 7.07 V
(d) 10 V, 7.07 V

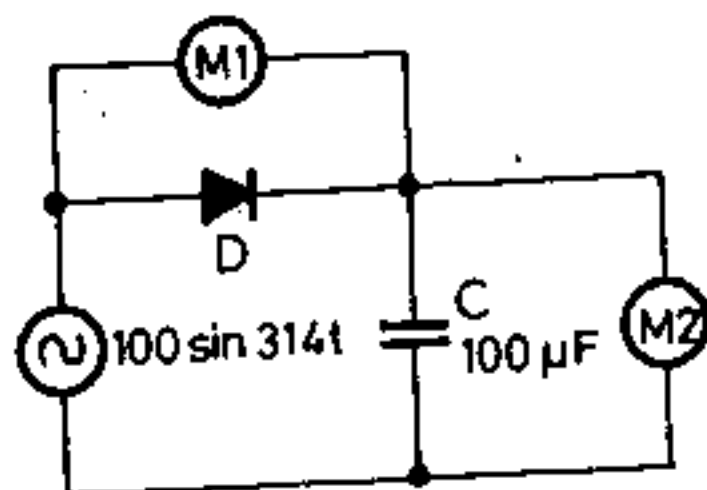


Fig. A-13

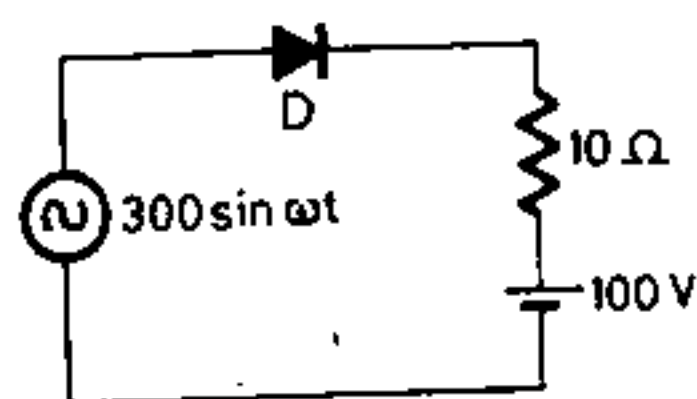


Fig. A-14

17. In Fig. A.14, PIV required for the diode is
(a) 300 V
(b) 100 V
(c) 200 V
(d) 400 V
18. A single-phase one-pulse diode rectifier is feeding an RL load with freewheeling diode across the load. For conduction angle β , the main diode and freewheeling diode would, respectively, conduct for
(a) $\pi, \pi - \beta$
(b) $\pi, \beta - \pi$
(c) β, π
(d) $\beta - \pi, \pi$
19. A single-phase full-bridge diode rectifier delivers a load current of 10 A, which is ripple free. Average and rms values of diode currents are respectively
(a) 10 A, 7.07 A
(b) 5 A, 10 A
(c) 5 A, 7.07 A
(d) 7.07 A, 5 A
20. A single-phase full-bridge diode rectifier delivers a constant load current of 10 A. Average and rms values of source current are respectively
(a) 5 A, 10 A
(b) 10 A, 10 A
(c) 5 A, 5 A
(d) 0, 10 A

21. A voltage $v = 4 \sin \omega t$ is applied to the terminals A and B of the circuit shown in Fig. A.15. The diodes are assumed to be ideal. The impedance offered by the circuit across the terminals A and B in kilo-ohms is

(a) 5 (b) 20
(c) 10 (d) none of these

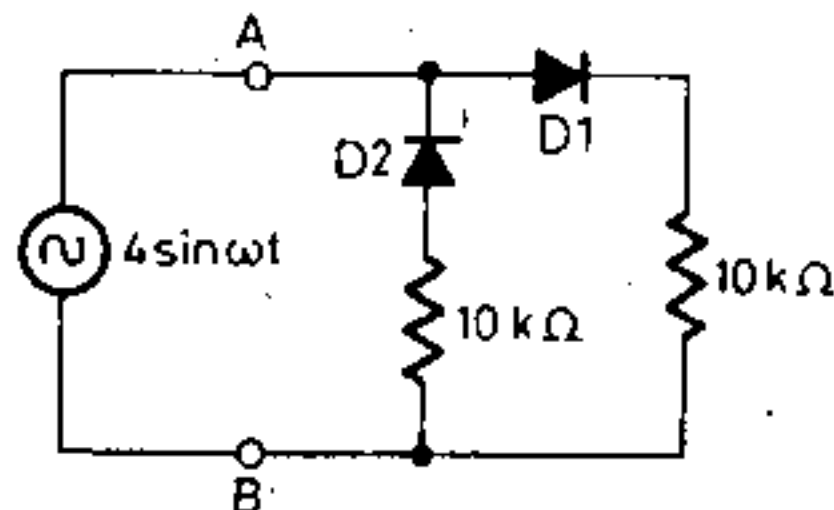


Fig. A-15

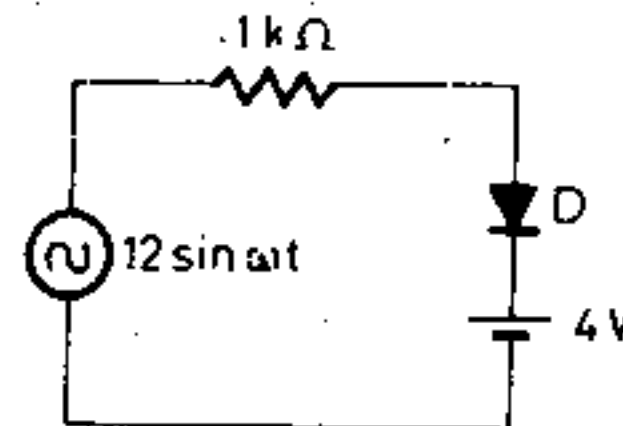


Fig. A-16

22. The peak current through the resistor of circuit of Fig. A.16, assuming the diodes to be ideal, is

(a) 12 mA (b) 4 mA
(c) 16 mA (d) 8 mA

23. In Fig. A.17, V1 and V2 are zero-centre PMMC voltmeters. When a sinusoidal signal is applied, V2 reads +20 V. The reading of the voltmeter V1 is volts.

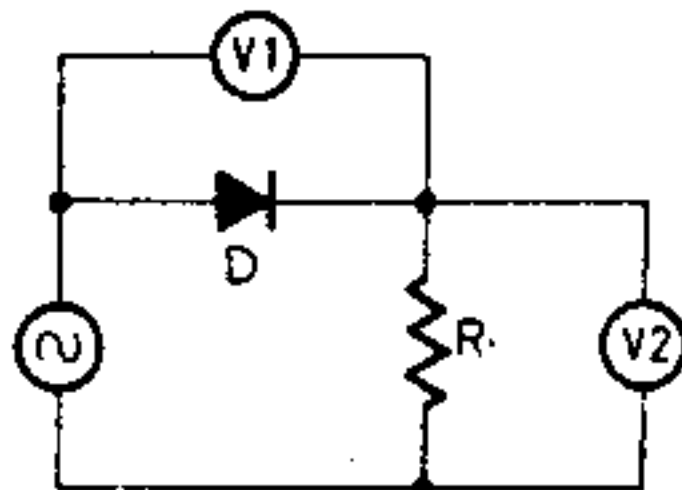


Fig. A-17

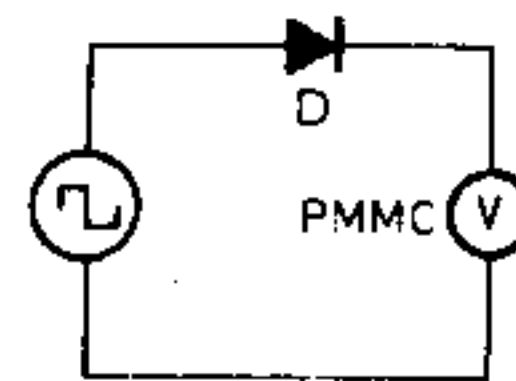


Fig. A-18

24. For a symmetrical square wave of 800 V peak to peak and for ideal diode, the voltmeter in Fig. A.18 will read

(a) 200 V (b) 400 V
(c) 800 V (d) zero

25. The circuit in Fig. A.19 shows zener-regulated dc power supply. The zener-diode is ideal. The minimum value of R_L down to which the output voltage remains constant is

(a) 27 Ω (b) 45 Ω
(c) 15 Ω (d) 24 Ω

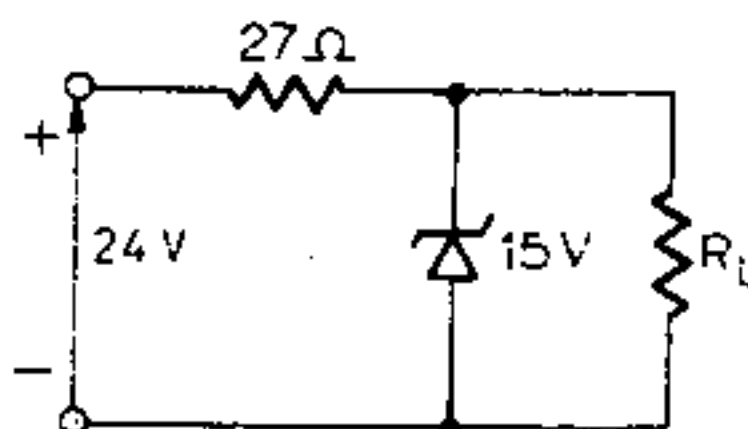


Fig. A-19

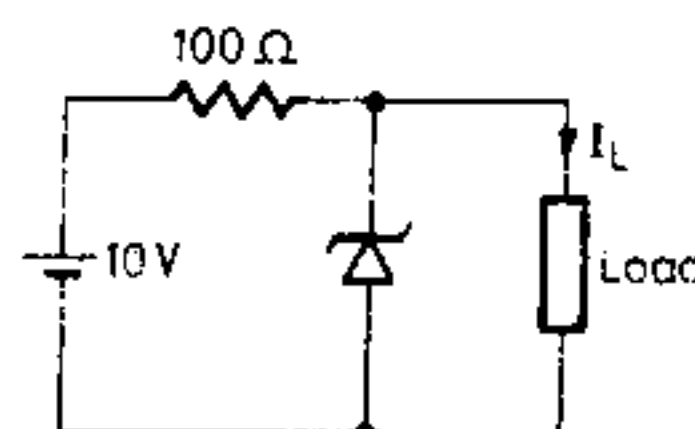


Fig. A-20

26. In the circuit of Fig. A.20, the 5V zener diode requires a minimum current of 10 mA. For obtaining a regulated output of 5 V, the maximum permissible load current I_L is mA and the minimum power rating of zener diode is W.

27. Fig. A.21 shows an electronic voltage regulator. The zener diode may be assumed to require a minimum current of 25 mA for satisfactory operation. The value of R required for satisfactory operation is ohms.

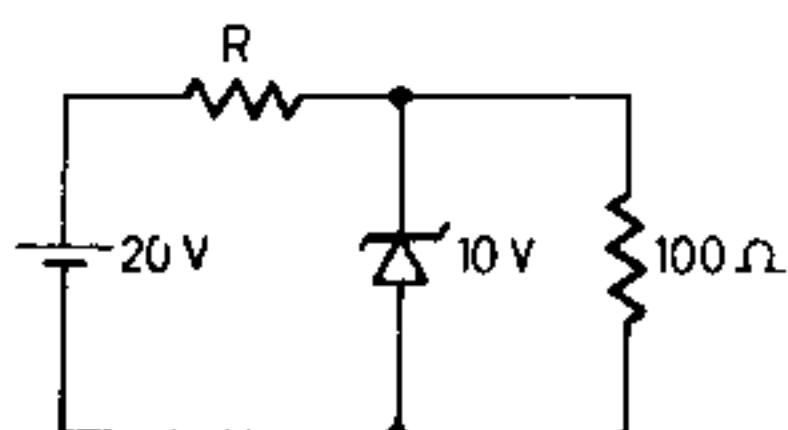


Fig. A-21

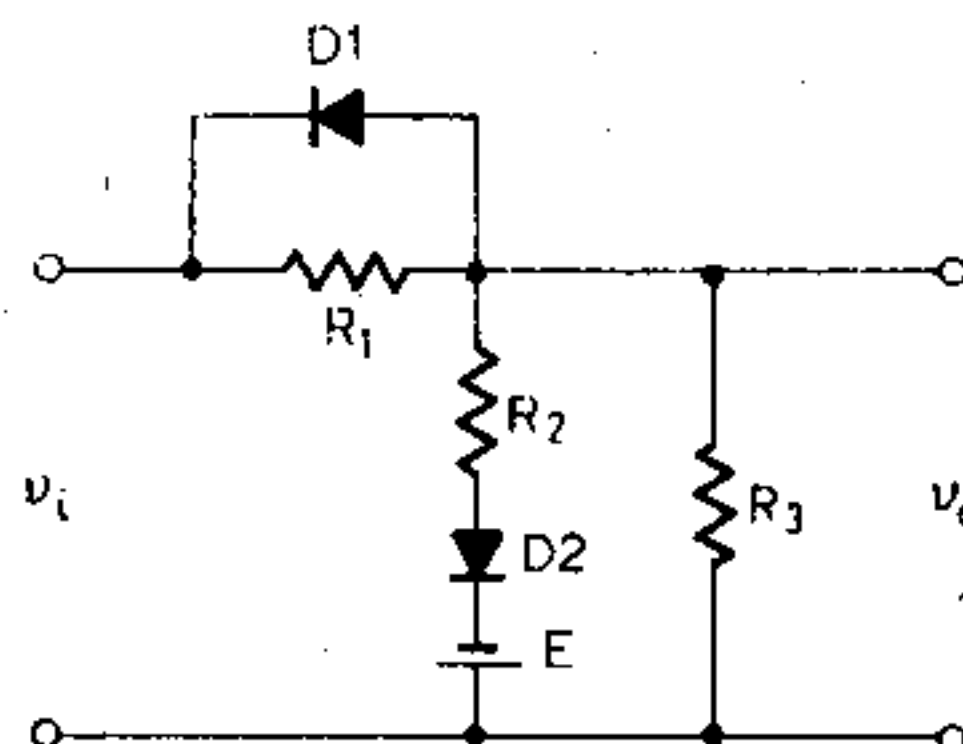


Fig. A-22

28. In the circuit of Fig. A.22, the diode states at the extremely large negative value of the input voltage v_i are
- (a) D1 off, D2 off (b) D1 on, D2 off
(c) D1 off, D2 on (d) D1 on, D2 on

ANSWERS

- | | | | | | |
|---------|----------------------|----------|---------|--------------|---------|
| 1. (a) | 2. (b) | 3. (a) | 4. (c) | 5. (b) | 6. (d) |
| 7. (d) | 8. (b) | 9. (b) | 10. (a) | 11. 10 mA, 0 | 12. (d) |
| 13. (c) | 14. (b) | 15. (b) | 16. (a) | 17. (d) | 18. (b) |
| 19. (c) | 20. (d) | 21. (c) | 22. (d) | 23. - 20V | 24. (a) |
| 25. (b) | 26. 40 mA and 0.05 W | 27. 80 Ω | 28. (b) | | |

THYRISTORS

- The number of $p-n$ junctions in a thyristor is
 - 1
 - 2
 - 3
 - 4.
- When anode is positive with respect to cathode in an SCR, the number of blocked $p-n$ junctions is
 - 1
 - 2
 - 3
 - 4
- When cathode is positive with respect to anode in an SCR, the number of blocked $p-n$ junctions is
 - 1
 - 2
 - 3
 - 4.
- In a thyristor, anode current is made up of
 - electrons only
 - electrons or holes
 - electrons and holes
 - holes only.
- A thyristor, when triggered, will change from forward blocking state to conduction state if its anode to cathode voltage is equal to
 - peak repetitive off-state forward voltage
 - peak working off-state forward voltage

- (c) peak working off-state reverse voltage
 - (d) peak non-repetitive off-state forward voltage.
6. An SCR can be brought to forward conducting state with gate-circuit open when the applied voltage exceeds
- (a) the forward breakover voltage
 - (b) reverse breakdown voltage
 - (c) 1.5 V
 - (d) peak non-repetitive off-state voltage.
7. A conducting thyristor can be turned-off by(decreasing/increasing) the anode current...(below/above) the ...(latching/holding) current and applying a ...(reverse/forward/zero) voltage across the SCR for a short interval.
8. In a thyristor, holding current is
- (a) more than latching current I_L
 - (b) less than I_L
 - (c) equal to I_L
 - (d) very small.
9. When a thyristor gets turned on, the gate drive
- (a) should not be removed as it will turn-off the SCR
 - (b) may or may not be removed
 - (c) should be removed
 - (d) should be removed in order to avoid increased losses and higher junction temperature.
10. For normal SCRs, turn-on time is
- (a) less than turn-off time, t_q
 - (b) more than t_q
 - (c) equal to t_q
 - (d) half of t_q .
11. The forward voltage drop during SCR on-state is 1.5 V. This voltage drop
- (a) remains constant and is independent of load current
 - (b) increases slightly with load current
 - (c) decreases slightly with load current
 - (d) varies linearly with load current.
12. During forward blocking state, a thyristor is associated with
- (a) large current, low voltage
 - (b) low current, large voltage
 - (c) medium current, large voltage
 - (d) low current, medium voltage.
13. In a thyristor, ratio of latching current to holding current is
- (a) 0.4
 - (b) 1.0
 - (c) 2.5
 - (d) 6.0.
14. On-state voltage drop across a thyristor used in a 250-V supply system is of the order of
- (a) 100-110 V
 - (b) 240-250 V
 - (c) 1 to 1.5 V
 - (d) 0.5 to 1 V.
15. Once SCR starts conducting a forward current, its gate loses control over
- (a) anode circuit voltage only
 - (b) anode circuit current only
 - (c) anode circuit voltage and current
 - (d) anode circuit voltage, current and time,

16. In a thyristors
 - (a) latching current I_L is associated with turn-off process and holding current I_H with turn-on process
 - (b) both I_L and I_H are associated with turn-off process
 - (c) I_H is associated with turn-off process and I_L with turn-on process
 - (d) both I_L and I_H are associated with turn-on process.
17. The SCR ratings, di/dt in A/ μ sec and dv/dt in V/ μ sec. may vary, respectively, between
 - (a) 20 to 500, 10 to 100
 - (b) both 20 to 500
 - (c) both 10 to 100
 - (d) 50 to 300, 20 to 500.
18. A thyristor can be termed as
 - (a) DC switch
 - (b) AC switch
 - (c) either (a) or (b)
 - (d) square-wave switch.
19. In a thyristor, the magnitude of anode current will
 - (a) increase if gate current is increased
 - (b) decrease if gate current is decreased
 - (c) increase if gate current is decreased
 - (d) not change with any variation in gate current.
20. Static $V-I$ characteristics of an SCR with different gate drives applied to the gate are indicated by
 - (a) $I_{g2} > I_{g1} > I_{g0}$
 - (b) $V_{g2} > V_{g1} > V_{g0}$
 - (c) $P_{g2} > P_{g1} > P_{g0}$
 - (d) either (a) or (b).
21. Turn-on time of an SCR can be reduced by using a
 - (a) rectangular pulse of high amplitude and narrow width
 - (b) rectangular pulse of low amplitude and wide width
 - (c) triangular pulse
 - (d) trapezoidal pulse.
22. Turn-on time of an SCR in series with RL circuit can be reduced by
 - (a) increasing circuit resistance R
 - (b) decreasing R
 - (c) increasing circuit inductance L
 - (d) decreasing L .
23. For an SCR with turn-on time of 5 microsecond, an ideal trigger pulse should have
 - (a) short rise time with pulse width = 3 μ sec
 - (b) long rise time with pulse width = 6 μ sec
 - (c) short rise time with pulse width = 6 μ sec
 - (d) long rise time with pulse width = 3 μ sec.
24. A forward voltage can be applied to an SCR after its
 - (a) anode current reduces to zero
 - (b) gate recovery time
 - (c) reverse recovery time
 - (d) anode voltage reduces to zero.
25. Turn-off time of an SCR is measured from the instant
 - (a) anode current becomes zero
 - (b) anode voltage becomes zero
 - (c) anode voltage and anode current become zero at the same time
 - (d) gate current becomes zero.
26. Turn-on time for an SCR is 10 μ sec. If an inductance is inserted in the anode circuit, then the turn-on time will be

Appendix

- (a) 10 μ sec (b) less than 10 μ sec
(c) more than 10 μ sec (d) about 10 μ sec.
27. In an SCR, anode current flows over a narrow region near the gate during
(a) delay time t_d (b) rise time t_r and spread time t_p
(c) t_d and t_p (d) t_d and t_r .
28. Gate characteristic of a thyristor
(a) is a straight line passing through origin
(b) is of the type $V_g = a + b I_g$
(c) is a curve between V_g and I_g
(d) has a spread between two curves of $V_g - I_g$.
29. The average on-state current for an SCR is 20 A for a conduction angle of 120° . Its average on-state current for 60° conduction angle would be
(a) 20 A (b) 10 A
(c) less than 20 A (d) 40 A.
30. The average on-state current for an SCR is 20 A for a resistive load. If an inductance of 5 mH is included in the load, then average on-state current would be
(a) more than 20 A (b) less than 20 A
(c) 15 A (d) 20 A.
31. Specification sheet for an SCR gives its maximum rms on-state current as 35 A. This rms rating for a conduction angle of 120° would be
(a) more than 35 A (b) less than 35 A
(c) 35 A (d) 52.5 A.
32. Surge current rating of an SCR specifies the maximum
(a) repetitive current with sine wave
(b) non-repetitive current with rectangular wave
(c) non-repetitive current with sine wave
(d) repetitive current with rectangular wave.
33. The di/dt rating of an SCR is specified for its
(a) decaying anode current (b) decaying gate current
(c) rising gate current (d) rising anode current.
34. For an SCR, dv/dt protection is achieved through the use of
(a) RL in series with SCR (b) RC across SCR
(c) L in series with SCR (d) RC in series with SCR.
35. For an SCR, di/dt protection is achieved through the use of
(a) R in series with SCR (b) RL in series with SCR
(c) L in series with SCR (d) L across SCR.
36. The function of snubber circuit connected across an SCR is to
(a) suppress dv/dt (b) increase dv/dt
(c) decrease dv/dt (d) keep transient overvoltage at a constant value.
37. The object of connecting resistance and capacitance across gate circuit is to protect the SCR gate against
(a) overvoltages (b) dv/dt
(c) noise signals (d) overcurrents.
38. Thermal resistance in SCRs has the units of..... ($W/^\circ C$, or ohms or $^\circ C/W$) and heat sinks are made from.....(copper/steel/aluminium).

39. Who string efficiency for n series/parallel connected SCRs is defined, in general, as

$$\frac{\text{.....}}{\text{.....} \times n}$$

40. Practical way of obtaining static voltage equalization in series-connected SCRs is by the use of

- (a) one resistor across the string
- (b) resistors of different values across each SCR
- (c) resistors of the same value across each SCR
- (d) one resistor in series with the string.

41. For series connected SCRs, dynamic equalizing circuit consists of

- (a) resistor R and capacitor C in series but with a diode D across C
- (b) series R and D circuit but with C across R
- (c) series R and C circuit but with D across R
- (d) series C and D circuit but with R across C .

42. For dynamic equalizing circuit used for series connected SCRs, the choice of C is based on

- (a) reverse recovery characteristic
- (b) turn-on characteristics
- (c) turn-off characteristics
- (d) rise-time characteristics.

43. In an UJT, with V_{BB} as the voltage across two base terminals, the emitter potential at peak point is given by

- (a) ηV_{BB}
- (b) ηV_D
- (c) $\eta V_{BB} + V_D$
- (d) $\eta V_D + V_{BB}$.

44. An UJT exhibits negative resistance region

- (a) before the peak point
- (b) between peak and valley points
- (c) after the valley point
- (d) both (a) and (c).

45. In an UJT, maximum value of charging resistance is associated with

- (a) peak point
- (b) valley point
- (c) any point between peak and valley points
- (d) after the valley point.

46. When an UJT is used for triggering an SCR, the waveshape of the voltage obtained from UJT circuit is a

- (a) sine wave
- (b) saw-tooth wave
- (c) trapezoidal wave
- (d) square wave.

47. For an UJT employed for the triggering of an SCR, stand-off ratio $\eta = 0.64$ and dc source voltage V_{BB} is 20 V. The UJT would trigger when the emitter voltage is

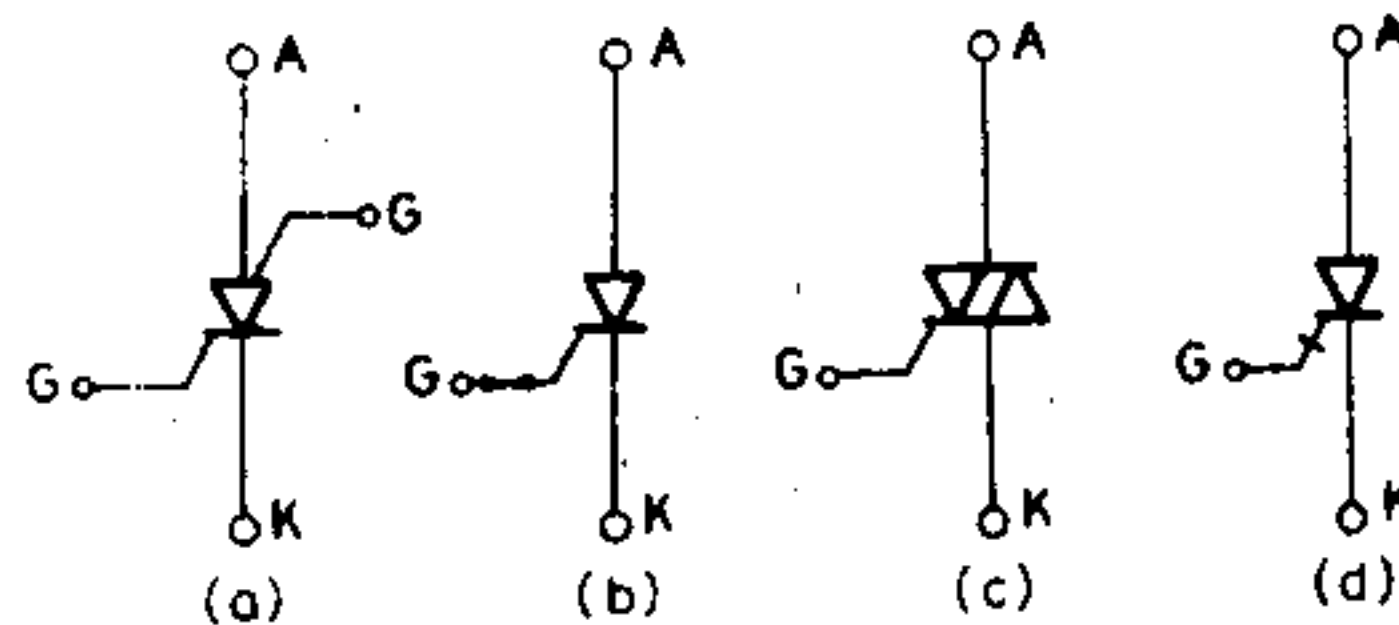
- (a) 12.8 V
- (b) 13.1 V
- (c) 10 V
- (d) 5 V.

48. An SCR can withstand a maximum junction temperature of 120°C with an ambient temperature of 75°C . If this SCR has thermal resistance from junction to ambient as 1.5°C/W , the maximum internal power dissipation allowed is

- (a) 30 W
- (b) 60 W
- (c) 80 W
- (d) 50 W.

49. In synchronized UJT triggering of an SCR, voltage v_c across capacitor reaches UJT threshold voltage thrice in each half cycle so that there are three firing pulses during each half cycle. The firing angle of the SCR can be controlled
- (a) once in each half cycle
 - (b) thrice in each half cycle
 - (c) twice in each half cycle
 - (d) four times in each half cycle.
50. The function of connecting a zener diode in an UJT circuit, used for the triggering of SCRs, is to
- (a) expedite the generation of triggering pulses
 - (b) delay the generation of triggering pulses
 - (c) provide a constant voltage to UJT to prevent erratic firing
 - (d) provide a variable voltage to UJT as the source voltage changes.
51. A metal oxide varistor (MOV) is used for protecting
- (a) gate circuit against overcurrents
 - (b) gate circuit against overvoltages
 - (c) anode circuit against overcurrents
 - (d) anode circuit against overvoltages.
52. The functions of connecting a resistor in series with gate-cathode circuit and a zener-diode across gate-cathode circuit are, respectively, to protect the gate circuit from
- (a) overvoltages, overcurrents
 - (b) overcurrents, overvoltages
 - (c) overcurrents, noise signals
 - (d) noise signals, overvoltages.
53. In a GTO, anode current begins to fall when gate current
- (a) is negative peak at time $t = 0$
 - (b) is negative peak at $t = \text{storage period } t_s$
 - (c) just begins to become negative at $t = 0$
 - (d) is negative peak at $t = (t_s + \text{fall time})$.
54. For a pulse transformer, the material used for its core and the possible turn-ratio from primary to secondary are, respectively,
- (a) ferrite ; 20 : 1
 - (b) laminated iron ; 1 : 1
 - (c) ferrite ; 1 : 1
 - (d) powdered iron ; 1 : 1.
55. The capacitance of a reverse biased junction of a thyristor is 20 picofarad. The charging current of this thyristor is 4 mA. The limiting value of $\frac{dv}{dt}$ is V/ μ s.
56. This question has five subquestions. Amongst the four alternatives provided, one or more are correct. Write down all the correct answers unambiguously.
- 56.1. SCR can be turned on by
- (a) applying anode voltage at a sufficiently fast rate
 - (b) applying sufficiently large anode voltage
 - (c) increasing the temperature of SCR to a sufficiently large value
 - (d) applying sufficiently large gate current.
- 56.2. During forward blocking of two series connected SCRs, a thyristor with
- (a) high leakage impedance shares lower voltage
 - (b) high leakage impedance shares higher voltage
 - (c) low leakage impedance shares higher voltage
 - (d) low leakage impedance shares lower voltage.

56.3. The circuit symbol for a GTO is



56.4. Thyristors *A* has rated gate current of 2 A and thyristor *B* a rated gate current of 100 mA.

- (a) *A* is a GTO and *B* is a conventional SCR
- (b) *B* is a GTO and *A* is a conventional SCR
- (c) *A* may operate as a transistor
- (d) *B* may operate as a transistor.

56.5. A resistor connected across the gate and cathode of an SCR increases its

- (a) dv/dt rating
- (b) holding current
- (c) noise immunity
- (d) turn-off time.

ANSWERS

- | | | | | | |
|--|----------------|----------------|---------------------|---------------------|---------|
| 1. (c) | 2. (a) | 3. (b) | 4. (c) | 5. (b) | 6. (a) |
| 7. decreasing, below, holding, reverse | | | 8. (b) | 9. (d) | 10. (a) |
| 11. (b) | 12. (b) | 13. (c) | 14. (c) | 15. (c) | 16. (c) |
| 17. (b) | 18. (a) | 19. (d) | 20. (a) | 21. (a) | 22. (d) |
| 23. (c) | 24. (b) | 25. (a) | 26. (c) | 27. (d) | 28. (d) |
| 29. (c) | 30. (a) | 31. (c) | 32. (c) | 33. (d) | 34. (b) |
| 35. (c) | 36. (a) | 37. (c) | 38. °C/W, Aluminium | | |
| 39. See the book | | 40. (c) | 41. (c) | 42. (a) | 43. (c) |
| 44. (b) | 45. (a) | 46. (b) | 47. (b) | 48. (a) | 49. (a) |
| 50. (c) | 51. (d) | 52. (b) | 53. (b) | 54. (c) | 55. 200 |
| 56.1 (a), (b), (c), (d) | 56.2. (b), (d) | 56.3. (b), (d) | 56.4 (a), (c) | 56.5. (a), (b), (c) | |

THYRISTOR COMMUTATION TECHNIQUES

1. For the circuit shown in Fig. A.23, the conduction time for thyristor in microseconds is

- (a) 0.393
- (b) 2.546
- (c) 25.133
- (d) 8.0

2. For the circuit in Fig. A.23, the capacitor voltage after SCR gets self-commutated is

- (a) 200 V
- (b) 400 V
- (c) 300 V
- (d) 100 V

3. For the circuit shown in Fig. A.23, the voltage across thyristor, after it is self-commutated is

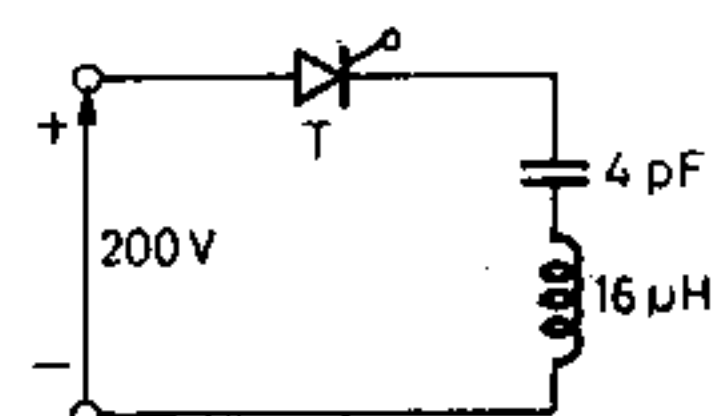


Fig. A-23

- (a) zero (b) -1.5 V
 (c) -200 V (d) -400 V
4. For the circuit in Fig. A.23, the peak thyristor current is
 (a) 100 A (b) 50 A
 (c) 400 A (d) 800 A
5. In the circuit of Fig. A.24, the maximum value of current through thyristors T1 and TA can respectively be

- (a) $\frac{V_s}{R}, \frac{V_s}{R} + V_s \sqrt{\frac{C}{L}}$
 (b) $\frac{V_s}{R} + V_s \sqrt{\frac{C}{L}}, V_s \sqrt{\frac{C}{L}}$
 (c) $V_s \sqrt{\frac{C}{L}}, \frac{V_s}{R}$
 (d) $\frac{V_s}{R}, V_s \sqrt{\frac{C}{L}}$

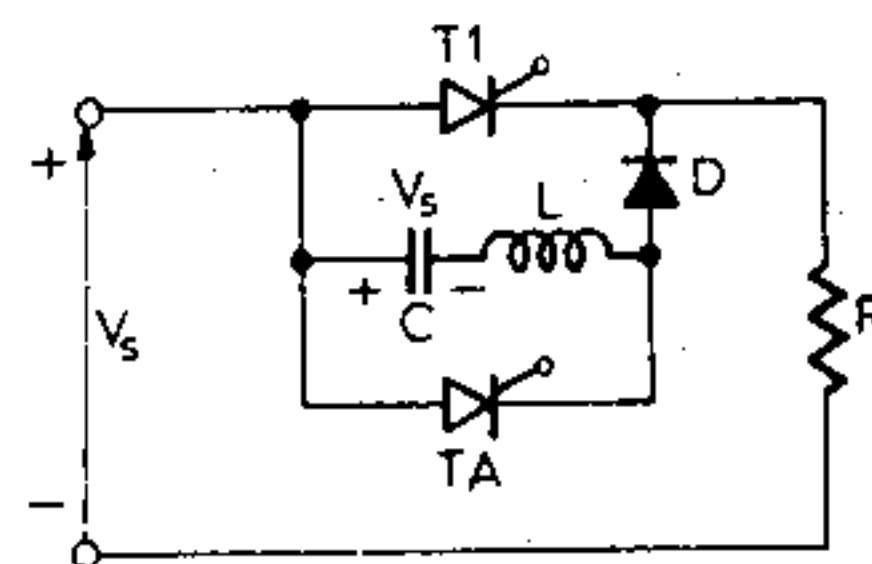


Fig. A-24

6. For the circuit shown in Fig. A.24, peak value of resonant current is twice the load current. In case $V_s = 200\text{ V}$, the magnitude of voltage across main thyristor, when it gets turned-off, is V.
7. For the circuit in Fig. A.24, the peak value of current through auxiliary SCR is twice that through the main SCR. In case $V_s = 100\text{ V}$, $C = 10\text{ }\mu\text{F}$ and load current $= 40\text{ A}$, the circuit turn-off time for main SCR is microsecond.
8. In the circuit of Fig. A.25, C is charged to V_s with polarity as shown. Tick the correct statement/statements from the following :
- (a) In order to turn-off T1, turn on T2
 (b) In order to turn-off T2, turn on T1
 (c) At the time of turn on of SCR, initial thyristor current is $V_s \left[\frac{2}{R_1} + \frac{1}{R_2} \right]$
 (d) At the time of turn on of SCR, initial thyristor current is $V_s \left[\frac{1}{R_1} + \frac{2}{R_2} \right]$

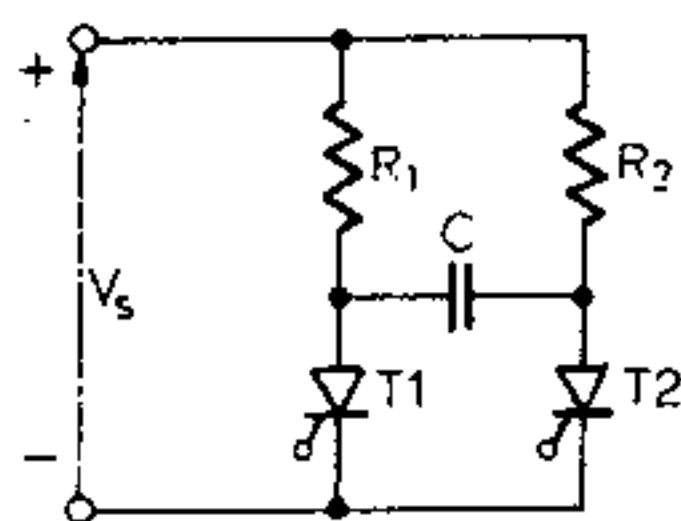


Fig. A-25

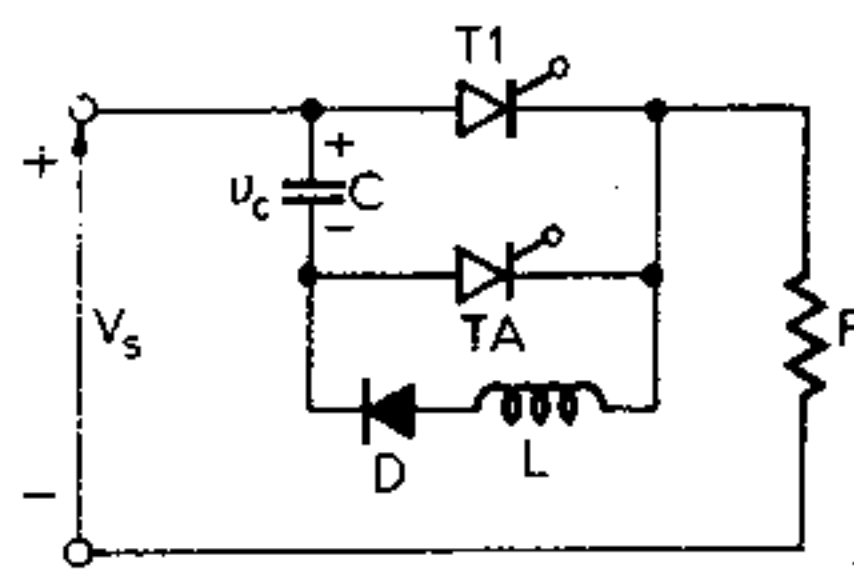


Fig. A-26

9. In the circuit shown in Fig. A.25, $R_1 = 50\text{ }\Omega$, $R_2 = 100\text{ }\Omega$ and $V_s = 100\text{ V}$. The possible peak values of current through T1 is A and through T2 is A.
10. In the circuit shown in Fig. A.26, $V_s = 200\text{ V}$, $C = 4\text{ }\mu\text{F}$, $L = 16\text{ }\mu\text{H}$ and $R = 20\text{ }\Omega$. The peak value of current through T1 and D can respectively be
 (a) $110\text{ A}, 100\text{ A}$
 (b) $10\text{ A}, 110\text{ A}$
 (c) $110\text{ A}, 10\text{ A}$
 (d) $100\text{ A}, 110\text{ A}$

11. In the circuit of Fig. A.26, and for the parameters given in Prob 10, the circuit turn-off time for main and auxiliary thyristors are respectively μs and μs .

12. In the circuit configuration shown in Fig. A.27, the circuit turn-off time for main thyristor is $34.657 \mu\text{s}$. The value of capacitor C required in this circuit is μF .

13. Match the two sides.

Type of commutation

Power Circuit Diagram

(A) Self-commutation

(B) Complementary commutation

(C) Impulse commutation

(D) Resonant-pulse commutation

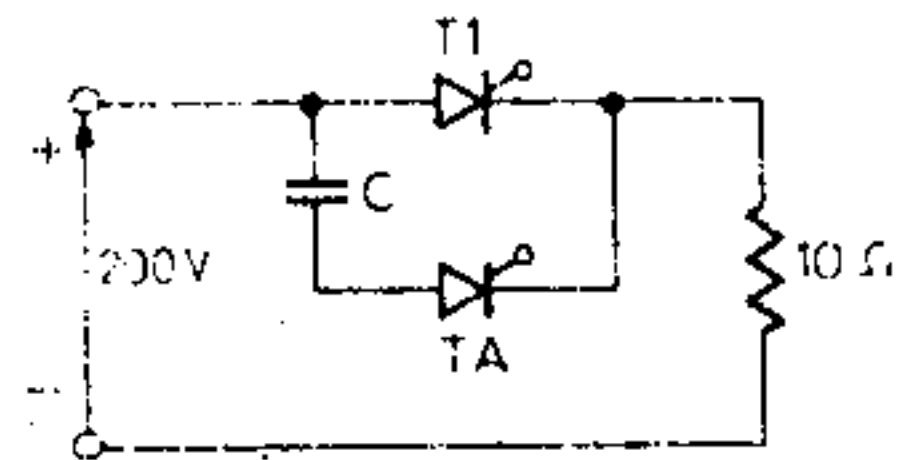
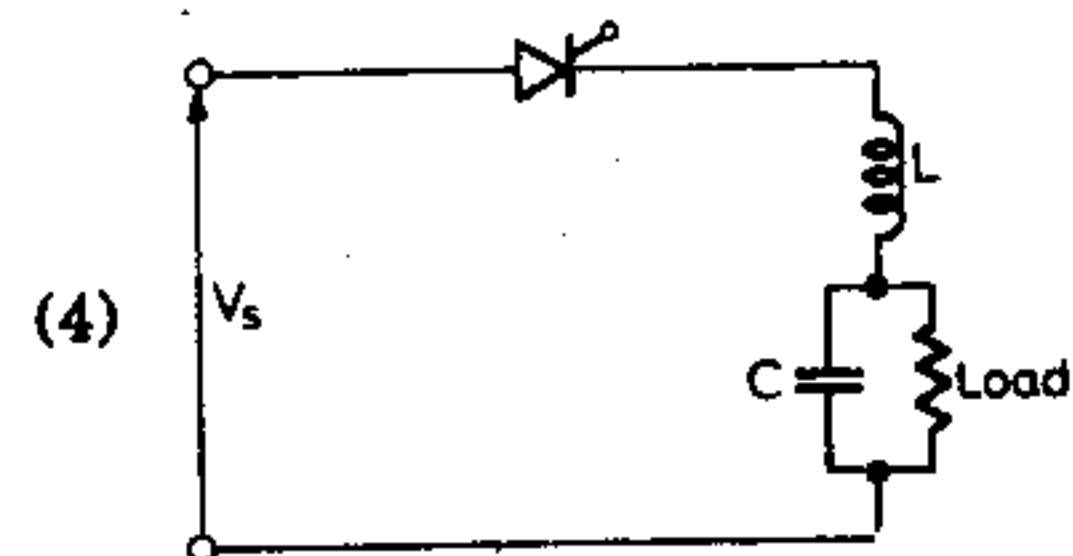
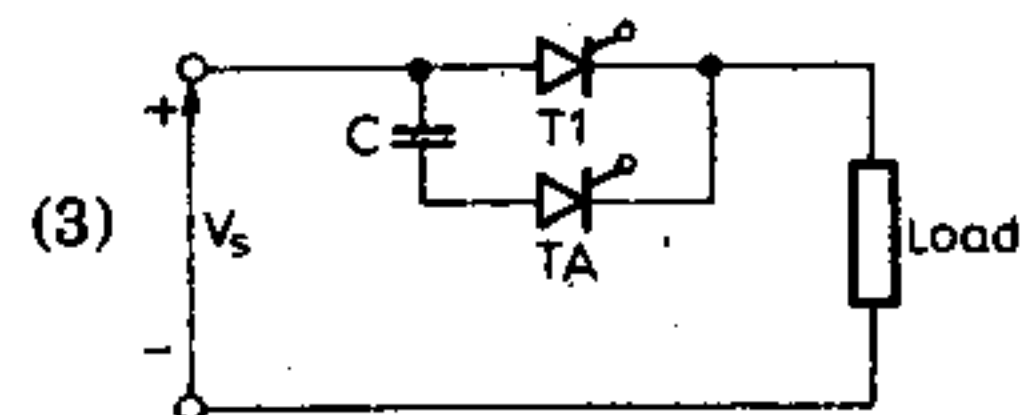
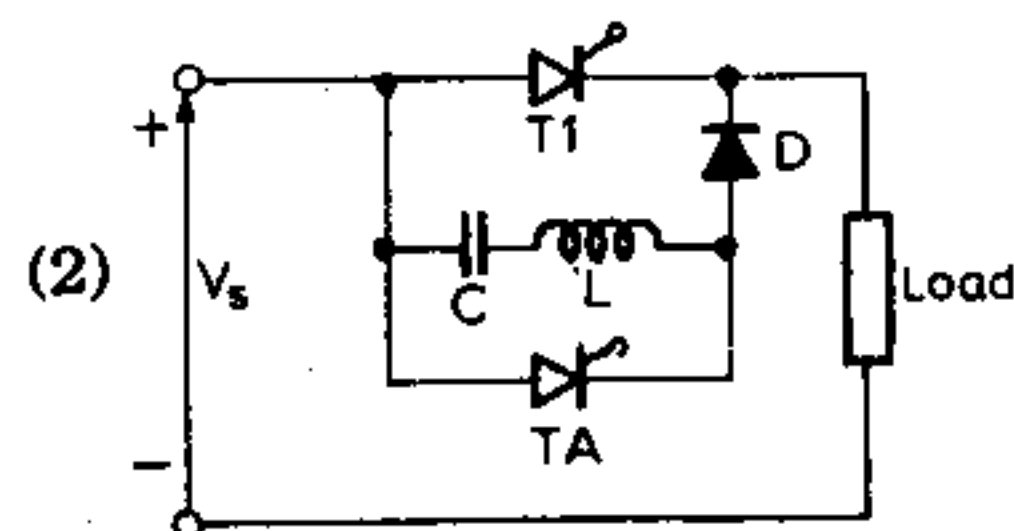
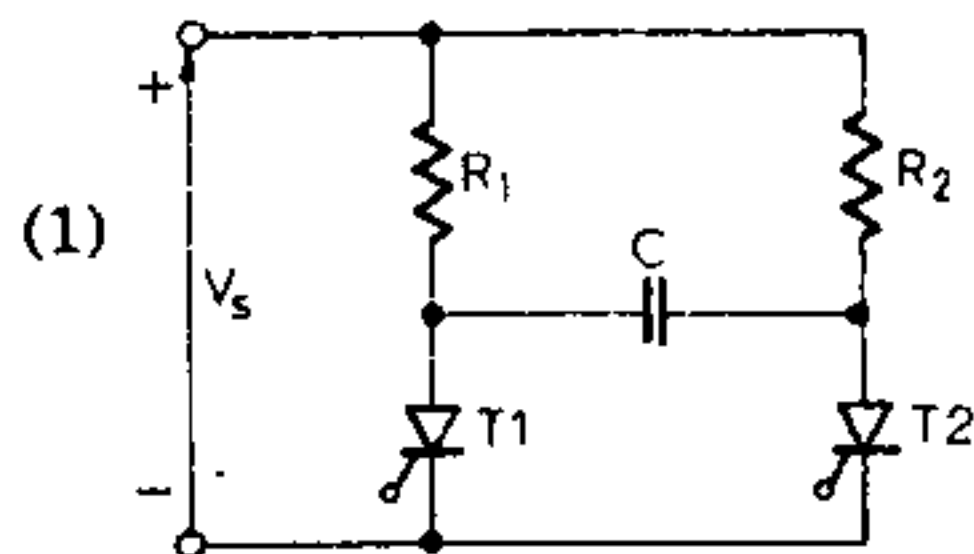


Fig. A-27



ANSWERS

- | | | | | | |
|----------|-------------|-----------|---------|----------------|------------|
| 1. (c) | 2. (b) | 3. (c) | 4. (a) | 5. (d) | 6. 173.2 V |
| 7. 21.65 | 8. (a), (c) | 9. 4A, 5A | 10. (a) | 11. 80, 12.566 | 12. 5 |
13. A $\rightarrow$ 4, B $\rightarrow$ 1, C $\rightarrow$ 3, D $\rightarrow$ 2.

PHASE CONTROLLED RECTIFIERS

1. A single-phase half-wave controlled rectifier has $400 \sin 314 t$ as the input voltage and R as the load. For a firing angle of 60° for the SCR, the average output voltage is
 (a) $400/\pi$ (b) $300/\pi$
 (c) $240/\pi$ (d) $200/\pi$
2. A single-phase one-pulse controlled circuit has resistance and counter emf load and $400 \sin 314 t$ as the source voltage. For a load counter emf of 200 V , the range of firing angle control is
 (a) 30° to 150° (b) 30° to 180°
 (c) 60° to 120° (d) 60° to 180°
3. In a single-phase half-wave circuit with RL load, and a freewheeling diode across the load, extinction angle β is more than π . For a firing angle α , the SCR and freewheeling diode would conduct, respectively, for
 (a) $\pi - \alpha, \beta$ (b) $\beta - \alpha, \pi - \alpha$
 (c) $\pi - \alpha, \beta - \pi$ (d) $\pi - \alpha, \pi - \beta$
4. In a single-phase one-pulse circuit with RL load and a freewheeling diode, extinction angle β is less than π . For a firing angle α , the SCR and freewheeling diode would, respectively, conduct for
 (a) $\beta - \alpha, 0^\circ$ (b) $\pi - \alpha, \pi - \beta$
 (c) $\alpha, \beta - \alpha$ (d) $\beta - \alpha, \alpha$
5. A single-phase full-wave mid-point thyristor converter uses a $230/200 \text{ V}$ transformer with center tap on the secondary side. The P.I.V. per thyristor is
 (a) 100 V (b) 141.4 V
 (c) 200 V (d) 282.8 V
6. A single-phase two-pulse bridge converter has an average output voltage and power output of 500 V and 10 kW respectively. The SCRs used in the two-pulse bridge converter are now re-employed to form a single-phase two-pulse mid-point converter. This new controlled converter would give, respectively, an average output voltage and power output of
 (a) $500 \text{ V}, 10 \text{ kW}$ (b) $250 \text{ V}, 5 \text{ kW}$
 (c) $250 \text{ V}, 10 \text{ kW}$ (d) $500 \text{ V}, 5 \text{ kW}$
7. In a single-phase full converter bridge, the average output voltage is given by
 (a) $\frac{1}{\pi} \int_{\alpha}^{\pi + \alpha} V_m \cos \theta \, d\theta$ (b) $\frac{1}{\pi} \int_0^{\alpha + \pi} V_m \cos \theta \cdot d\theta$
 (c) $\frac{1}{\pi} \int_{\alpha - (\pi/2)}^{\alpha + (\pi/2)} V_m \cos \theta \cdot d\theta$ (d) $\frac{1}{\pi} \int_{(\pi/2) - \alpha}^{(\pi/2) + \alpha} V_m \cos \theta \cdot d\theta$
8. In a single-phase semiconverter, the average output voltage is given by
 (a) $\frac{1}{\pi} \int_{\alpha}^{\pi} V_m \cos \theta \cdot d\theta$ (b) $\frac{1}{\pi} \int_{(\pi/2) - \alpha}^{(\pi/2) + \alpha} V_m \cos \theta \cdot d\theta$
 (c) $\frac{1}{\pi} \int_{\alpha - (\pi/2)}^{\pi/2} V_m \cos \theta \cdot d\theta$ (d) $\frac{1}{\pi} \int_{\alpha - (\pi/2)}^{\pi} V_m \cos \theta \cdot d\theta$
9. In a single-phase full converter, for continuous conduction, each pair of SCRs conduct for

- (a) $\pi - \alpha$ (b) π
(c) α (d) $\pi + \alpha$.
10. In a single-phase full converter, for discontinuous load current and extinction angle $\beta > \pi$, each SCR conducts for
(a) α (b) $\beta - \alpha$
(c) β (d) $\alpha + \beta$
11. In a single-phase semi-converter, for continuous conduction, each SCR conducts for
(a) α (b) π
(c) $\alpha + \pi$ (d) $\pi - \alpha$
12. In a single-phase semiconverter, for discontinuous conduction and extinction angle $\beta > \pi$, each SCR conducts for
(a) $\pi - \alpha$ (b) $\beta - \alpha$
(c) α (d) β .
13. In a single-phase semiconverter, for discontinuous conduction and extinction angle $\beta < \pi$, each SCR conducts for
(a) $\pi - \alpha$ (b) $\beta - \alpha$
(c) α (d) β
14. In a single-phase semiconverter, for continuous conduction, freewheeling diode conducts for
(a) α (b) $\pi - \alpha$
(c) π (d) $\pi + \alpha$
15. In a single-phase semiconverter, with discontinuous conduction and extinction angle $\beta > \pi$, freewheeling diode conducts for
(a) α (b) $\beta - \pi$
(c) $\pi + \alpha$ (d) β .
16. In a single-phase semiconverter, with discontinuous conduction and extinction angle $\beta < \pi$, freewheeling diode conducts for
(a) α (b) $\pi - \beta$
(c) $\beta - \pi$ (d) zero degree.
17. In a single-phase full converter, if α and β are firing and extinction angles respectively, then the load current is
(a) discontinuous if $(\beta - \alpha) < \pi$ (b) discontinuous if $(\beta - \alpha) > \pi$
(c) discontinuous if $(\beta - \alpha) = \pi$ (d) continuous if $(\beta - \alpha) < \pi$.
18. In a single-phase full converter with resistive load and for a firing angle α , the load current is zero and non-zero, respectively, for
(a) $\alpha, \pi - \alpha$ (b) $\pi - \alpha, \alpha$
(c) $\alpha, \pi + \alpha$ (d) α, π .
19. In a single-phase semiconverter with resistive load and for a firing angle α , each SCR and freewheeling diode conduct, respectively, for
(a) $\alpha, 0^\circ$ (b) $\pi - \alpha, \alpha$
(c) $\pi + \alpha, \alpha$ (d) $\pi - \alpha, 0^\circ$.
20. In controlled rectifiers, the nature of load current, i.e. whether load current is continuous or discontinuous
(a) does not depend on type of load and firing angle delay
(b) depends both on the type of load and firing angle delay
(c) depends only on the type of load
(d) depends only on the firing angle delay.

21. In a single-phase full converter, if output voltage has peak and average values of 325 V and 133 V respectively, then the firing angle is
(a) 40° (b) 140°
(c) 50° (d) 130° .
22. In a single-phase semiconverter, if output voltage has peak and average values of 325 and 133 V respectively, the firing angle is
(a) 40° (b) 140°
(c) 73.40° (d) 80° .
23. Each diode of a 3-phase half-wave diode rectifier conducts for
(a) 60° (b) 120°
(c) 180° (d) 90° .
24. Each diode of a 3-phase, 6-pulse bridge diode rectifier conducts for
(a) 60° (b) 120°
(c) 180° (d) 90° .
25. In a 3-phase half-wave diode rectifier, if per phase input voltage is 200 V, then the average output voltage is
(a) 233.91 V (b) 116.95 V
(c) 202.56 V (d) 101.28 V.
26. In a 3-phase half-wave diode rectifier, the ratio of average output voltage to per-phase maximum ac voltage is
(a) 0.955 (b) 0.827
(c) 1.654 (d) 1.169.
27. For a 3-phase, six-pulse diode rectifier, the average output voltage in terms of maximum value of line voltage V_m is
(a) $\frac{3\sqrt{2}}{\pi} V_m$ (b) $\frac{3 V_m}{\pi}$
(c) $\frac{3\sqrt{3}}{2\pi} V_m$ (d) $\frac{3\sqrt{3}}{\pi} V_m$.
28. In a 3-phase half-wave rectifier, dc output voltage is 230 V. The peak inverse voltage across each diode is
(a) 481.7 V (b) 460 V
(c) 345 V (d) 230 V.
29. In a 3-phase full-wave diode rectifier, the peak inverse voltage in terms of average output voltage is
(a) 1.571 (b) 0.955
(c) 1.047 (d) 2.094.
30. In a 3-phase half-wave diode rectifier, if V_m is the maximum value of per phase voltage, then each diode is subjected to a peak inverse voltage of
(a) V_m (b) $\sqrt{3} V_m$
(c) $2V_m$ (d) $3V_m$.
31. In a 3-phase full-wave diode rectifier, if V_m is the maximum value of line voltage, then each diode is subjected to a peak inverse voltage of
(a) V_m (b) $\sqrt{3} V_m$
(c) $2V_m$ (d) $3V_m$.
32. In a 3-phase full-wave diode rectifier, if V is the per phase input voltage, then average output voltage is given by

- (a) 0.955 V (b) 1.35 V
(c) 2.34 V (d) 3 V.
33. A converter which can operate in both 3-pulse and 6-pulse modes is a
(a) 1-phase full converter (b) 3-phase half-wave converter
(c) 3-phase semiconverter (d) 3-phase full converter.
34. In a 3-phase semi-converter, for firing angle less than or equal to 60° , each thyristor and diode conduct, respectively, for
(a) $60^\circ, 60^\circ$ (b) $90^\circ, 30^\circ$
(c) $120^\circ, 120^\circ$ (d) $180^\circ, 180^\circ$.
35. In a 3-phase semiconverter, for firing angle less than or equal to 60° , freewheeling diode conducts for
(a) 30° (b) 60°
(c) 90° (d) zero degree
36. In a 3-phase semiconverter, for a firing angle equal to 90° and for continuous conduction, each SCR and diode conduct, respectively, for
(a) $30^\circ, 60^\circ$ (b) $60^\circ, 30^\circ$
(c) $60^\circ, 60^\circ$ (d) $30^\circ, 30^\circ$.
37. In a 3-phase semiconverter, for a firing angle equal to 90° and for continuous conduction, freewheeling diode conducts for
(a) 30° (b) 60°
(c) 90° (d) zero degree.
38. In a 3-phase semiconverter, for firing angle equal to 120° and extinction angle equal to 110° , each SCR and diode conduct, respectively, for
(a) $30^\circ, 60^\circ$ (b) $60^\circ, 60^\circ$
(c) $90^\circ, 30^\circ$ (d) $110^\circ, 30^\circ$.
39. In a 3-phase semiconverter, for firing angle equal to 120° and extinction angle equal to 110° , freewheeling diode conducts for
(a) 10° (b) 30°
(c) 50° (d) 110° .
40. In a 3-phase semiconverter, for firing angle equal to 120° and extinction angle equal to 100° , none of the bridge elements conduct for
(a) 10° (b) 20°
(c) 30° (d) 60° .
41. A 3-phase semiconverter can work as
(a) converter for $\alpha = 0^\circ$ to 180° (b) converter for $\alpha = 0^\circ$ to 90°
(c) inverter for $\alpha = 90^\circ$ to 180° (d) inverter for $\alpha = 0^\circ$ to 90° .
42. In a 3-phase semiconverter, the three SCRs are triggered at an interval of
(a) 60° (b) 90°
(c) 120° (d) 180° .
43. In a 3-phase full converter, the six SCRs are fired at an interval of
(a) 30° (b) 60°
(c) 90° (d) 120° .
44. In a 3-phase full converter, three SCRs pertaining to one group are fired at an interval of
(a) 30° (b) 60°
(c) 90° (d) 120° .

45. For a single-phase phase-controlled rectifier, with a freewheeling diode across the load,
- (a) the instantaneous output voltage v_0 is always positive
 - (b) v_0 may be positive or zero
 - (c) v_0 may be positive, zero or negative
 - (d) v_0 is always zero or negative.
46. The frequency of the ripple in the output voltage of a 3-phase semiconverter depends upon
- (a) firing angle and load resistance
 - (b) firing angle and load inductance
 - (c) the load circuit parameters
 - (d) firing angle and the supply frequency.
47. In a single-phase full converter, if load current is I and ripple free, then average thyristor current is
- (a) $\frac{1}{2} I$
 - (b) $\frac{1}{3} I$
 - (c) $\frac{1}{4} I$
 - (d) I .
48. In a 3-phase full converter, if load current is I and ripple free, then average thyristor current is
- (a) $\frac{1}{2} I$
 - (b) $\frac{1}{3} I$
 - (c) $\frac{1}{4} I$
 - (d) I .
49. The effect of source inductance on the performance of single-phase and three-phase full converters is to
- (a) reduce the ripples in the load current
 - (b) make discontinuous current as continuous
 - (c) reduce the output voltage
 - (d) increase the load voltage.
50. In a single-phase full converter, the number of SCRs conducting during overlap is
- (a) 1
 - (b) 2
 - (c) 3
 - (d) 4.
51. In a single-phase full converter, the output voltage during overlap is equal to
- (a) zero
 - (b) source voltage
 - (c) source voltage minus the inductance drop
 - (d) inductance drop.
52. In a 3-phase full converter, the output voltage during overlap is equal to
- (a) zero
 - (b) source voltage
 - (c) source voltage minus the inductance drop
 - (d) average value of the conducting-phase voltages.
53. The total number of SCRs conducting simultaneously in 3-phase full converter with overlap considered has the sequence of
- (a) 3, 3, 2, 2
 - (b) 3, 3, 3, 2
 - (c) 3, 2, 3, 2
 - (d) 2, 2, 2, 3.

54. A 3-phase full converter has an average output voltage of 200 V for zero degree firing angle and for resistive load. For a firing angle of 90° , the output voltage would be
- (a) zero
 - (b) 50 V
 - (c) 100 V
 - (d) 26.8 V.
55. A four quadrant operation requires
- (a) two full converters in series
 - (b) two full converters connected back to back
 - (c) two full converters connected in parallel
 - (d) two semiconverters connected back to back.
56. In a 3-phase full converter, the output voltage pulsates at a frequency equal to
- (a) supply frequency, f
 - (b) $2f$
 - (c) $3f$
 - (d) $6f$.
57. In circulating-current type of dual converter, the nature of voltage across reactor is
- (a) alternating
 - (b) pulsating
 - (c) direct
 - (d) triangular.
58. The peak inverse voltage in ac to dc converter systems is highest in
- (a) single-phase full wave mid-point converter
 - (b) single-phase full converter
 - (c) 3-phase bridge converter
 - (d) 3-phase half-wave converter.
59. The three-phase ac to dc converter which requires neutral point connection is
- (a) 3-phase semiconverter
 - (b) 3-phase full converter
 - (c) 3-phase half-wave converter
 - (d) 3-phase full converter with diodes.
60. This question has five subquestions. Amongst the four alternatives provided, one or more are correct. Write down all the correct answers unambiguously.
- 60.1. A 3-phase full converter can function as
- (a) converter for $\alpha = 0^\circ$ to 180°
 - (b) converter for $\alpha = 0^\circ$ to 90°
 - (c) inverter for $\alpha = 90^\circ$ to 180°
 - (d) inverter for $\alpha = 0^\circ$ to 90° .
- 60.2. In a 3-phase full converter, the number of SCRs working during overlap is
- (a) 1 from positive group, 1 from negative group
 - (b) two from positive group, 1 from negative group
 - (c) 1 from positive group, 2 from negative group
 - (d) 2 from positive group, 2 from negative group.
- 60.3. In a 3-phase semiconverter, frequency of the ripple in the output voltage may be
- (a) 3 times the supply frequency f for firing angle $\alpha < 60^\circ$
 - (b) $3f$ for $\alpha > 60^\circ$
 - (c) $6f$ for $\alpha < 60^\circ$
 - (d) $6f$ for $\alpha > 60^\circ$.
- 60.4. In a dual converter, converters 1 and 2 work as under :
- (a) 1 as rectifier, 2 as inverter
 - (b) 1 as inverter, 2 as rectifier
 - (c) both as rectifiers
 - (d) both as inverters.

60.5. For the same ac voltage and load impedance which of the following statements about rectifiers are correct ?

- (a) The average load current in a full-wave rectifier is twice that in a half wave rectifier
- (b) The average load current in a full-wave rectifier is π times that in a half-wave rectifier
- (c) Half-wave rectifier will have bigger sized transformer compared to full-wave rectifier
- (d) Half-wave rectifier will have a smaller sized transformer compared to a full-wave rectifier.

61. This question contains four sub-questions each having two columns. Match every item on the left side with the most appropriate item on the right side. Give your answers as shown in the example below :

Example. (A)-(P)

(B)-(Q)

(C)-(R)

(D)-(S).

61.1. Single-phase full converter with 50 Hz supply

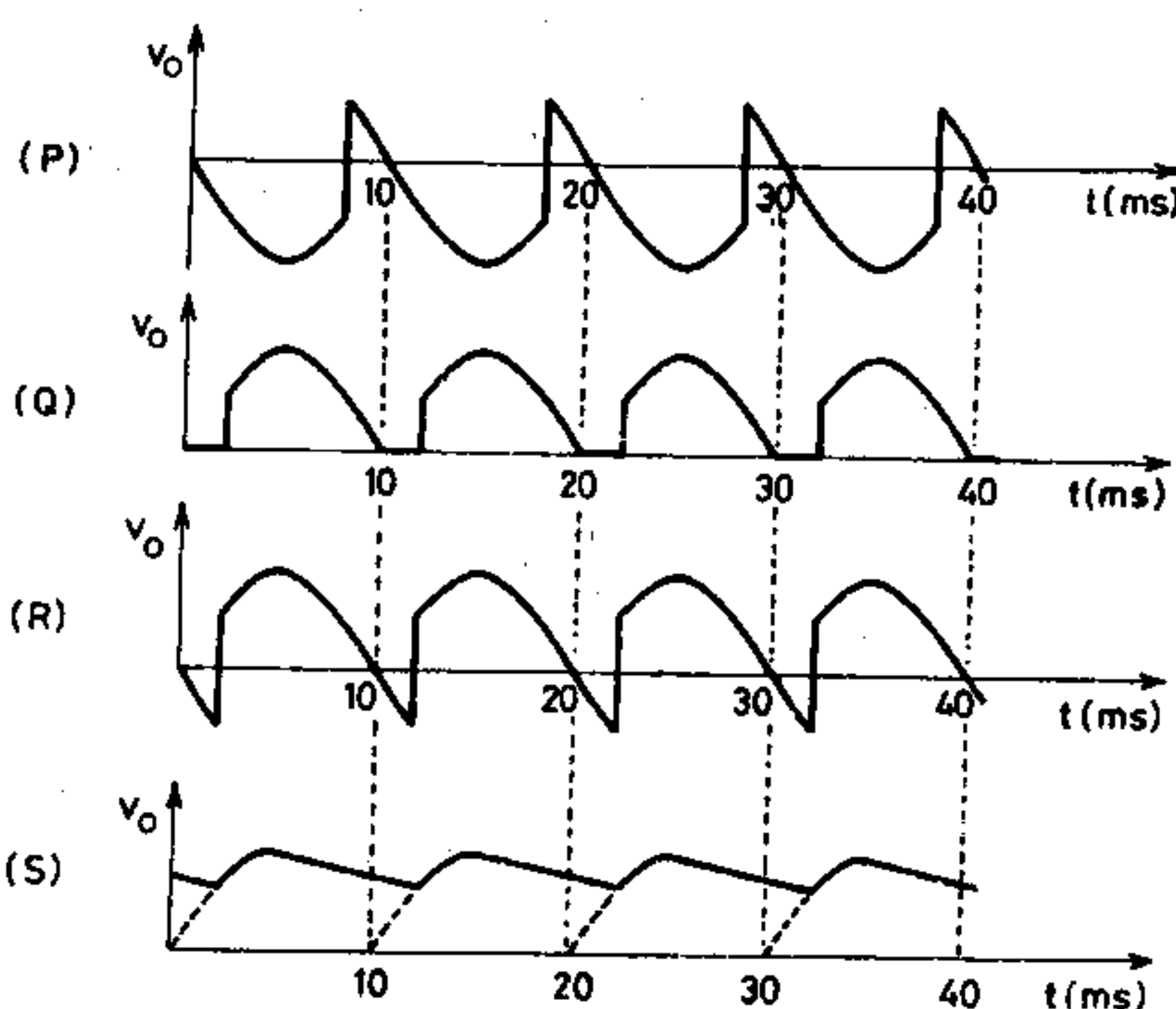
Output voltage waveforms

(A) R-C (parallel) load

(B) Continuous conduction mode

(C) Resistance load

(D) Inverter mode.



61.2. Controlled rectifiers

Output voltage expressions

(A) 1-phase full converter

$$(P) \frac{3 V_m}{2\pi} (1 + \cos \alpha)$$

(B) 1-phase semiconverter

$$(Q) \frac{2 V_m}{\pi} \cos \alpha$$

(C) 3-phase half-wave converter

$$(R) \frac{V_m}{\pi} (1 + \cos \alpha)$$

(D) 3-phase semiconverter

$$(S) \frac{3\sqrt{3}}{2\pi} V_m \cos \alpha$$

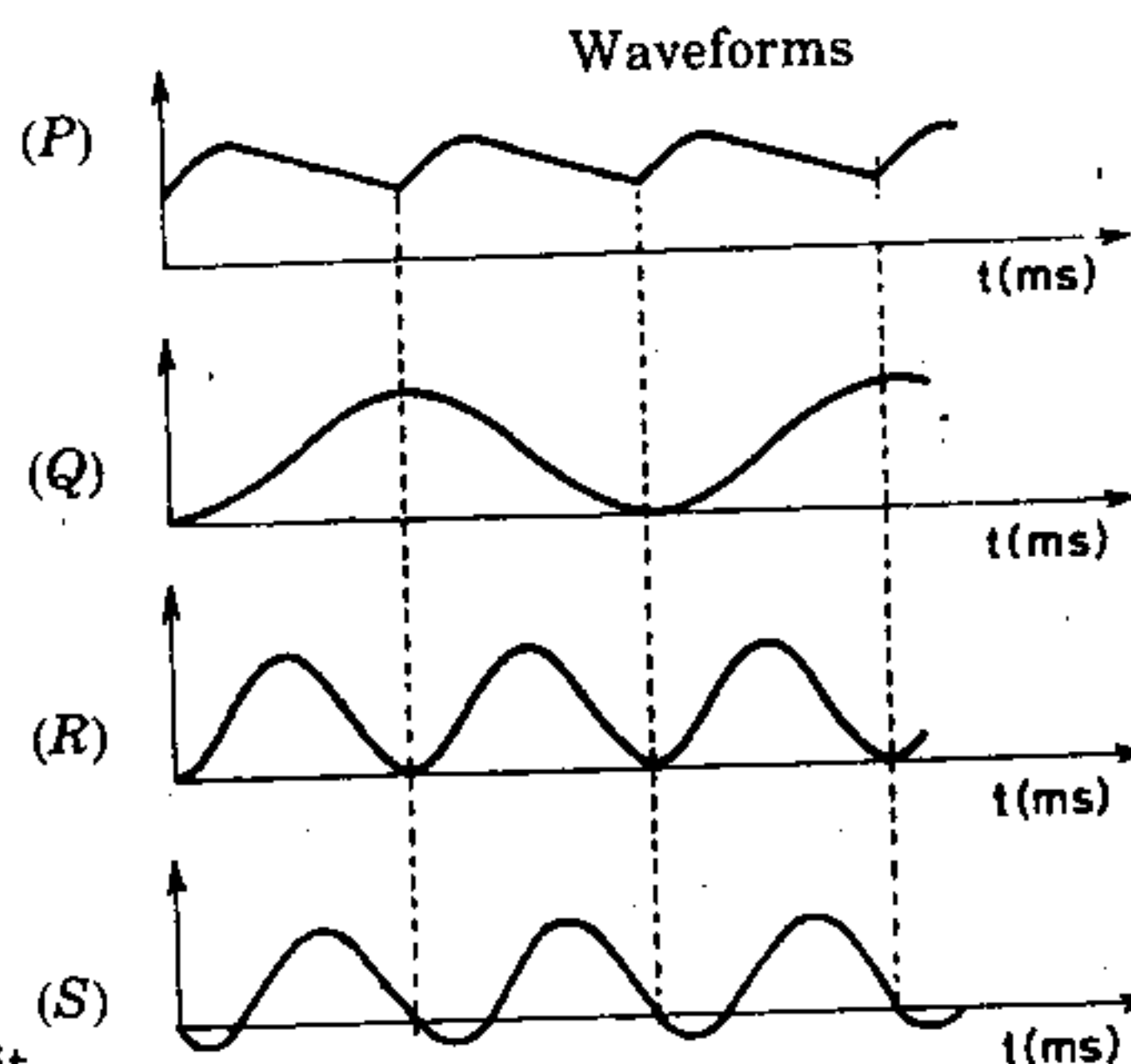
61.3. 50-Hz system measurements :

(A) Voltage across an R - C (parallel) load connected through a full-wave bridge

(B) Instantaneous power consumed by a resistor.

(C) Output voltage of a positive clamped circuit.

(D) Instantaneous power consumed by an R - L circuit.

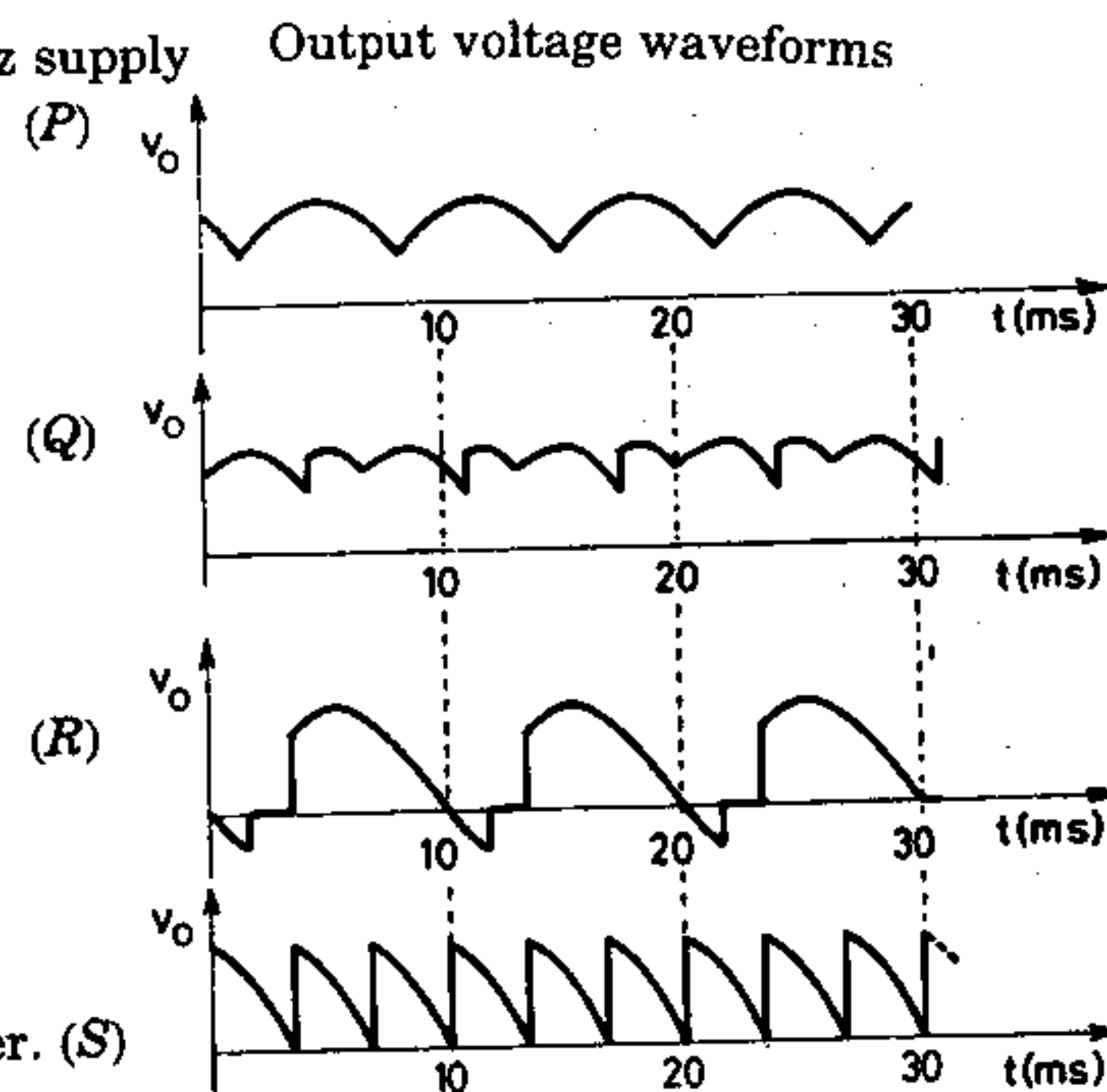
**61.4. Controlled rectifiers with 50 Hz supply**

(A) 1-phase full converter with source inductance

(B) 3-phase full converter

(C) 3-phase semiconverter

(D) 3-phase half-wave converter.

**ANSWERS**

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (a) | 3. (c) | 4. (a) | 5. (d) | 6. (b) |
| 7. (c) | 8. (c) | 9. (b) | 10. (b) | 11. (d) | 12. (a) |
| 13. (b) | 14. (a) | 15. (b) | 16. (d) | 17. (a) | 18. (a) |
| 19. (d) | 20. (b) | 21. (c) | 22. (c) | 23. (b) | 24. (b) |
| 25. (a) | 26. (b) | 27. (b) | 28. (a) | 29. (c) | 30. (b) |
| 31. (a) | 32. (c) | 33. (c) | 34. (c) | 35. (d) | 36. (c) |
| 37. (a) | 38. (b) | 39. (c) | 40. (b) | 41. (a) | 42. (c) |
| 43. (b) | 44. (d) | 45. (b) | 46. (d) | 47. (a) | 48. (b) |
| 49. (c) | 50. (d) | 51. (a) | 52. (d) | 53. (c) | 54. (d) |
| 55. (b) | 56. (d) | 57. (a) | 58. (a) | 59. (c) | |

- 60.1. (b), (c) 60.2. (b), (c) 60.3. (b), (c) 60.4. (a), (b) 60.5. (a), (d)
 61.1. (A)-(S) ; (B)-(R) ; (C)-(Q) ; (D)-(P) 61.2. (A)-(Q) ; (B)-(R) ; (C)-(S) ; (D)-(P)
 61.3. (A)-(P) ; (B)-(R) ; (C)-(Q) ; (D)-(S) 61.4. (A)-(R) ; (B)-(S) ; (C)-(Q) ; (D)-(P).

CHOPPERS

- In dc choppers, if T_{on} is the on-period and f is the chopping frequency, then output voltage in terms of input voltage V_s is given by
 (a) $V_s \cdot T_{on}/f$ (b) $V_s \cdot f/T_{on}$
 (c) $V_s/f \cdot T_{on}$ (d) $V_s \cdot f \cdot T_{on}$
- In dc choppers, the waveforms for input and output voltages are respectively
 (a) discontinuous, continuous (b) both continuous
 (c) both discontinuous (d) continuous, discontinuous.
- In PWM method of controlling the average output voltage in a chopper, the on-time is(varied/kept constant) but the chopping frequency is....(varied/kept constant).
- In FM method of controlling the average output voltage in a chopper, chopping period is....(varied/kept constant) but on-time is....(varied/kept constant) or off-time is....(varied/kept constant).
- For type-A chopper, V_s is the source voltage, R is the load resistance and α is the duty cycle. The average output voltage and current for this chopper are respectively
 (a) $\alpha V_s, \alpha \cdot (V_s/R)$ (b) $(1 - \alpha) V_s, (1 - \alpha) V_s/R$
 (c) $V_s/\alpha, V_s/\alpha R$ (d) $V_s/(1 - \alpha), V_s/(1 - \alpha)R$
- A chopper has V_s as the source voltage, R as the load resistance and α as the duty cycle. For this chopper, rms value of output voltage is
 (a) αV_s (b) $\sqrt{\alpha} \cdot V_s$
 (c) $V_s/\sqrt{\alpha}$ (d) $\sqrt{1 - \alpha} \cdot V_s$
- For a chopper, V_s is the source voltage. R is the load resistance and α is the duty cycle. Rms and average values of thyristor currents for this chopper are
 (a) $\alpha \cdot \frac{V_s}{R}, \sqrt{\alpha} \cdot \frac{V_s}{R}$ (b) $\sqrt{\alpha} \cdot \frac{V_s}{R}, \sqrt{\alpha} \cdot \frac{V_s}{R}$
 (c) $\sqrt{\alpha} \cdot \frac{V_s}{R}, \alpha \cdot \frac{V_s}{R}$ (d) $\sqrt{1 - \alpha} \cdot \frac{V_s}{R}, (1 - \alpha) V_s/R$
- In dc choppers, per unit ripple is maximum when duty cycle α is
 (a) 0.2 (b) 0.5
 (c) 0.7 (d) 0.9.
- In the circuit shown in Fig. A-28, $L = 5 \mu\text{H}$ and $C = 20 \mu\text{F}$. C is initially charged to 200 V. After the switch S is closed at $t = 0$, the maximum value of current and the time at which it reaches this value are, respectively,
 (a) 400 A, 15.707 μs (b) 50 A, 30 μs
 (c) 100 A, 62.828 μs (d) 400 A, 31.414 μs
- A voltage commutated chopper has the following parameters :

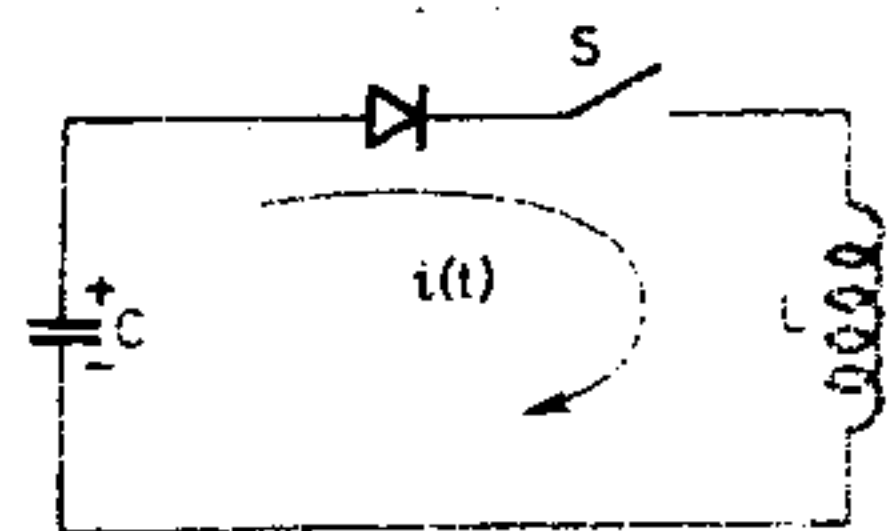


Fig. A-28

$V_s = 200$ V, Load circuit parameter : $1\ \Omega$, 2 mH , 50 V
 Commutation circuit parameters, $L = 25\ \mu\text{H}$, $C = 50\ \mu\text{F}$
 $T_{on} = 500\ \mu\text{s}$, $T = 2000\ \mu\text{s}$

For a constant load current of 100 A , the effective on period and peak current through the main thyristor are respectively

- (a) $1000\ \mu\text{s}$, 200 A (b) $700\ \mu\text{s}$, 382.8 A
 (c) $700\ \mu\text{s}$, 282.8 A (d) $1000\ \mu\text{s}$, 382.8 A .

11. For the voltage-commutated chopper of Prob. 10, the turn-off times for main and auxiliary thyristors are, respectively,

- (a) $120\ \mu\text{s}$, $60\ \mu\text{s}$ (b) $100\ \mu\text{s}$, $0.5\ \mu\text{s}$
 (c) $120\ \mu\text{s}$, $55\ \mu\text{s}$ (d) $100\ \mu\text{s}$, $55.54\ \mu\text{s}$.

12. In the current-commutated chopper shown in Fig. A-29, thyristor T1 is conducting a load current. When thyristor TA is turned off with capacitor polarity as shown the capacitor current i_c would flow through....(thyristor T1/diode D1), because.....

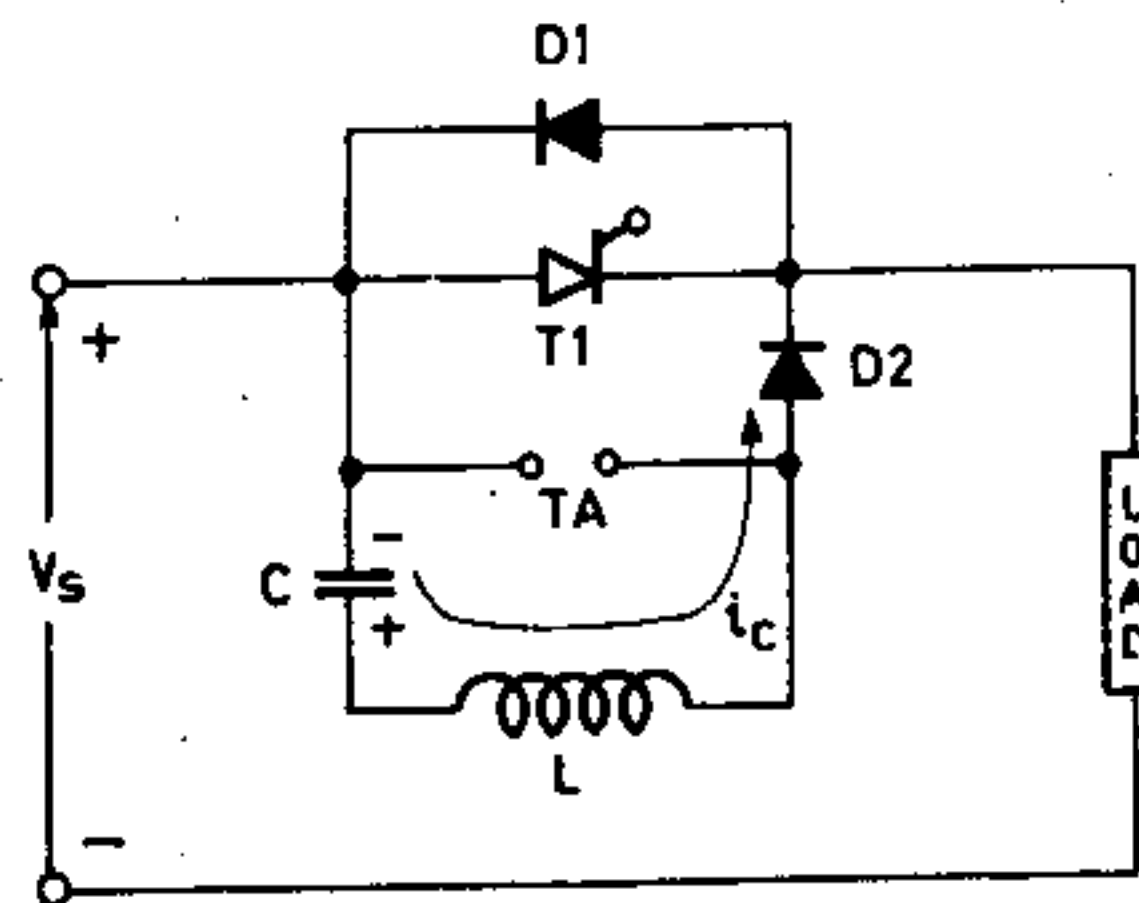


Fig. A-29

13. A load commutated chopper, fed from 200 V dc source, has a constant load current of 50 A . For a duty cycle of 0.4 and a chopping frequency of 2 kHz , the value of commutating capacitor and the turn-off time for one thyristor pair are respectively

- (a) $25\ \mu\text{F}$, $50\ \mu\text{s}$ (b) $50\ \mu\text{F}$, $50\ \mu\text{s}$
 (c) $25\ \mu\text{F}$, $25\ \mu\text{s}$ (d) $50\ \mu\text{F}$, $25\ \mu\text{s}$.

14. A dc battery is charged from a constant dc source of 200 V through a chopper. The dc battery is to be charged from its internal emf of 90 to 120 V . The battery has internal resistance of $1\ \Omega$. For a constant charging current of 10 A , the range of duty cycle is.....to.....

15. For type-A chopper ; V_s , R , I_0 and α are respectively the dc source voltage, load resistance, constant load current and duty cycle. For this chopper, average and rms values of freewheeling diode currents are

- (a) αI_0 , $\sqrt{\alpha} \cdot I_0$ (b) $(1 - \alpha) I_0$, $\sqrt{1 - \alpha} \cdot I_0$
 (c) $\alpha \cdot V_s/R$, $\sqrt{\alpha} \cdot V_s/R$ (d) $(1 - \alpha) I_0$, $\sqrt{\alpha} \cdot I_0$.

16. A step-up chopper has V_s as the source voltage and α as the duty cycle. The output voltage for this chopper is given by

- (a) $V_s(1 + \alpha)$ (b) $V_s/(1 - \alpha)$
 (c) $V_s(1 - \alpha)$ (d) $V_s/(1 + \alpha)$.

17. A dc chopper is fed from 100 V dc. Its load voltage consists of rectangular pulses of duration 1 msec in an overall cycle time of 3 msec. The average output voltage and ripple factor for this chopper are respectively

(a) 25 V, 1 (b) 50 V, 1 (c) 33.33 V, $\sqrt{2}$ (d) 33.33 V, 1

18. When a series LC circuit is connected to a dc supply of V volts through a thyristor, then the peak current through thyristor is

(a) $V \cdot \sqrt{LC}$ (b) $V/\sqrt{CL}$
(c) $V \cdot \sqrt{C/L}$ (d) $V \cdot \sqrt{L/C}$

19. For the arrangement shown in Fig. A-30, the circuit is initially in steady state with thyristor T off. After thyristor T is turned on, the peak thyristor current would be

(a) 2 A (b) 22 A
(c) 40 A (d) 42 A.

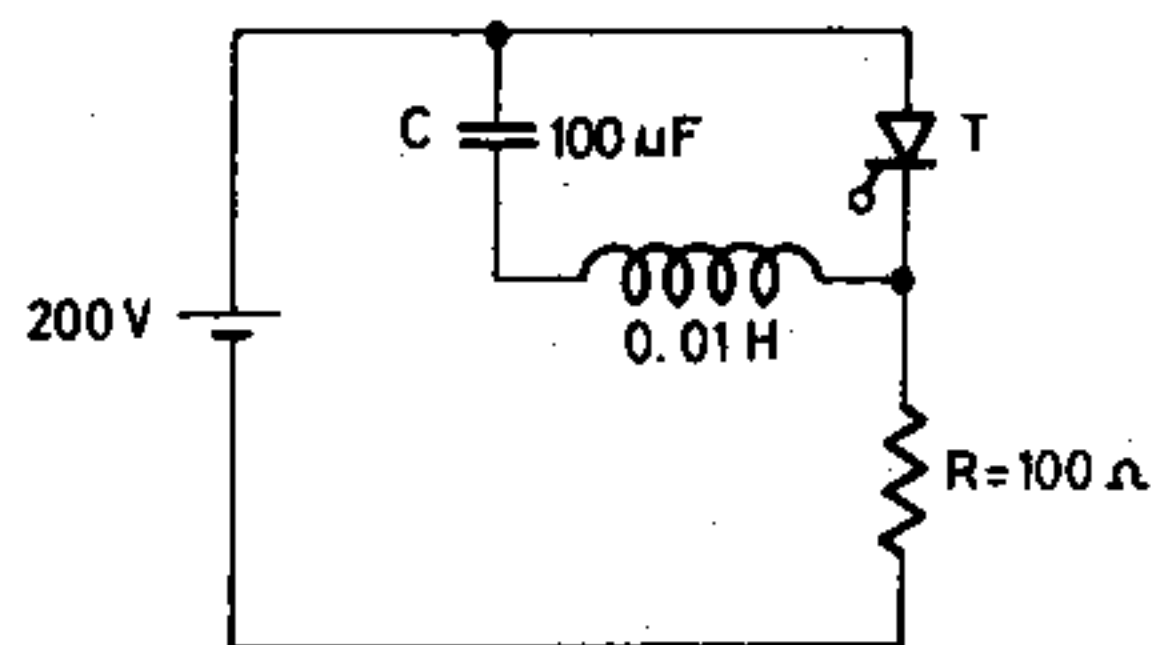


Fig. A-30

20. In type-A chopper, source voltage is 100 V dc, on-period = 100 μ s, off-period = 150 μ s and load RLE consists of $R = 2 \Omega$, $L = 5$ mH, $E = 10$ V. For continuous conduction, average output voltage and average output current for this chopper are respectively:

(a) 40 V, 15 A (b) 66.66 V, 28.33 A
(c) 60 V, 25 A (d) 40 V, 20 A.

21. Refer to the circuit in Fig. A-31. The maximum current in the main SCR M can be

(a) 200 A (b) 170.7 A
(c) 141.4 A (d) 70.7 A.

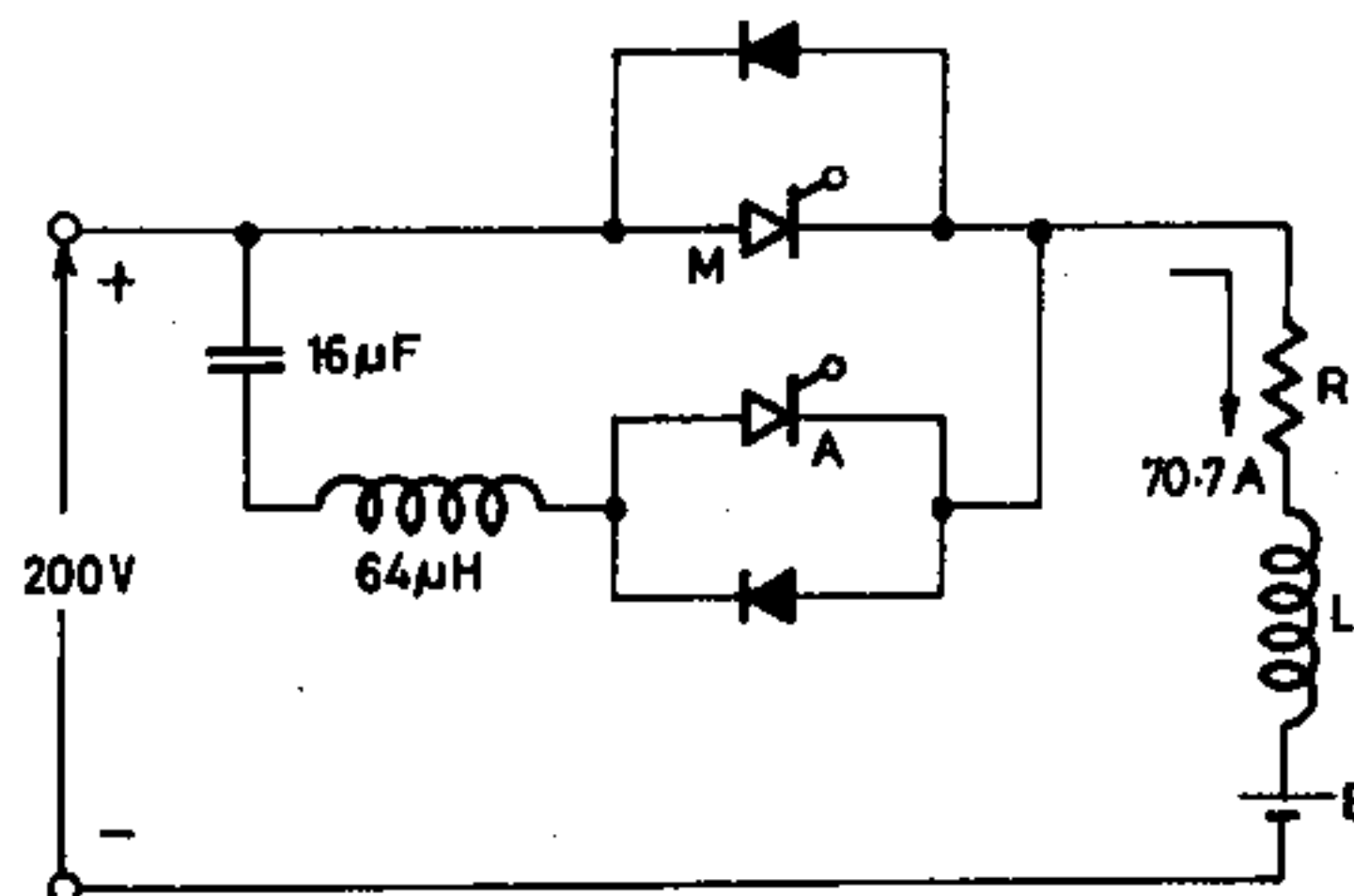


Fig. A-31

22. Refer to the circuit in Fig. A-31. The maximum turn-off time of the main SCR M to ensure its proper commutation, in μ s, is

(a) 2π (b) 4π (c) 6π (d) 8π

23. This question has two subquestions. Amongst the four alternatives provided, one or more are correct. Write down all the correct answers unambiguously.

- 23.1. In dc choppers, if T is the chopping period, then output voltage can be controlled by PWM by varying

(a) T keeping T_{on} constant (b) T_{on} keeping T constant
(c) T_{off} keeping T constant (d) T keeping T_{off} constant.

23.2. In dc choppers, for chopping period T , the output voltage can be controlled by FM by varying

- (a) T keeping T_{on} constant (b) T keeping T_{off} constant
 (c) T_{on} keeping T constant (d) T_{off} keeping T constant.

24. This question contains two subquestions, each having two columns. Match every item on the left side with the most appropriate item on the right side. Give your answers as shown in the example below :

Example. (A)-(P) ; (B)-(Q) ; (C)-(R) ; (D)-(S).

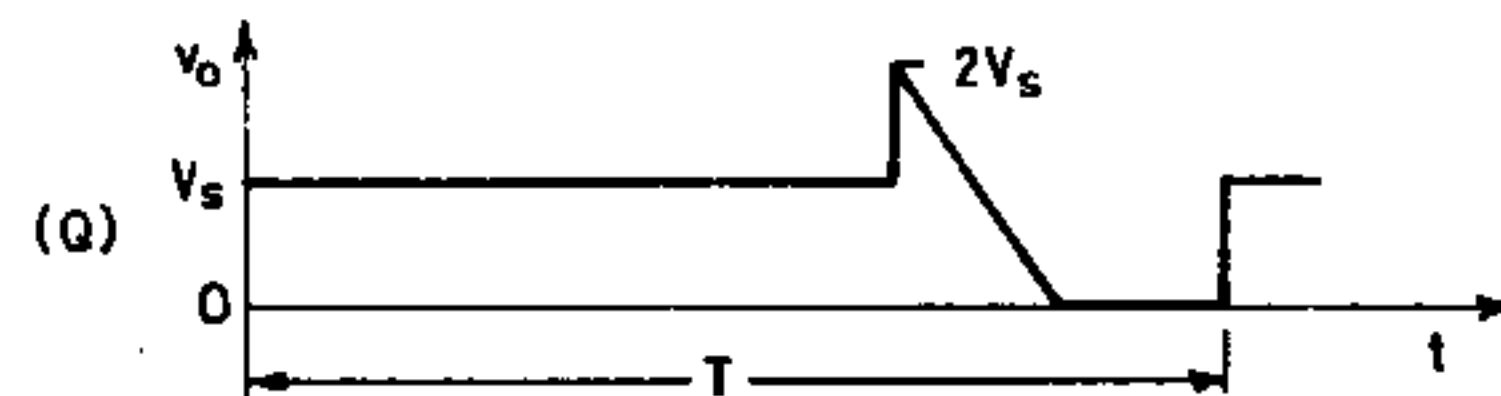
24.1. Chopper configurations

Output Voltage Waveforms

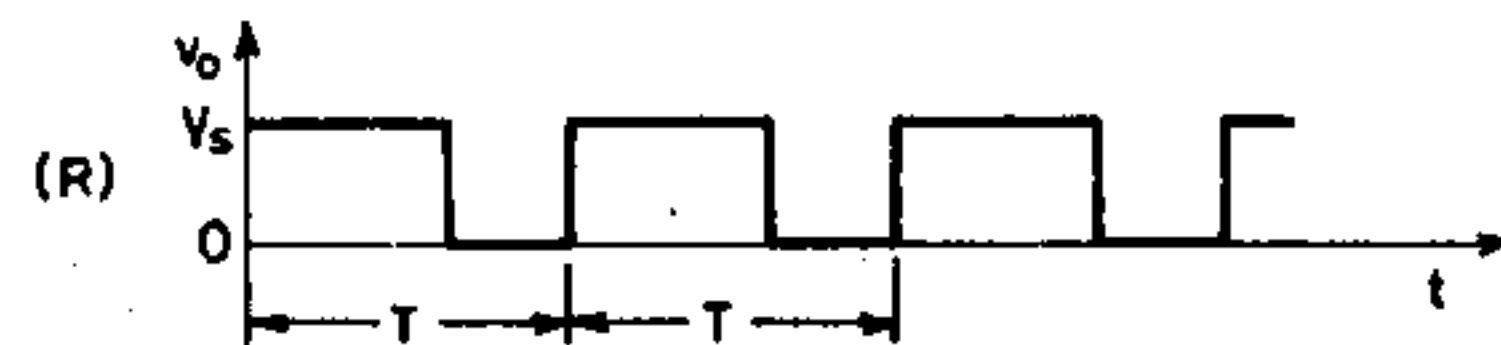
(A) Voltage-commutated chopper



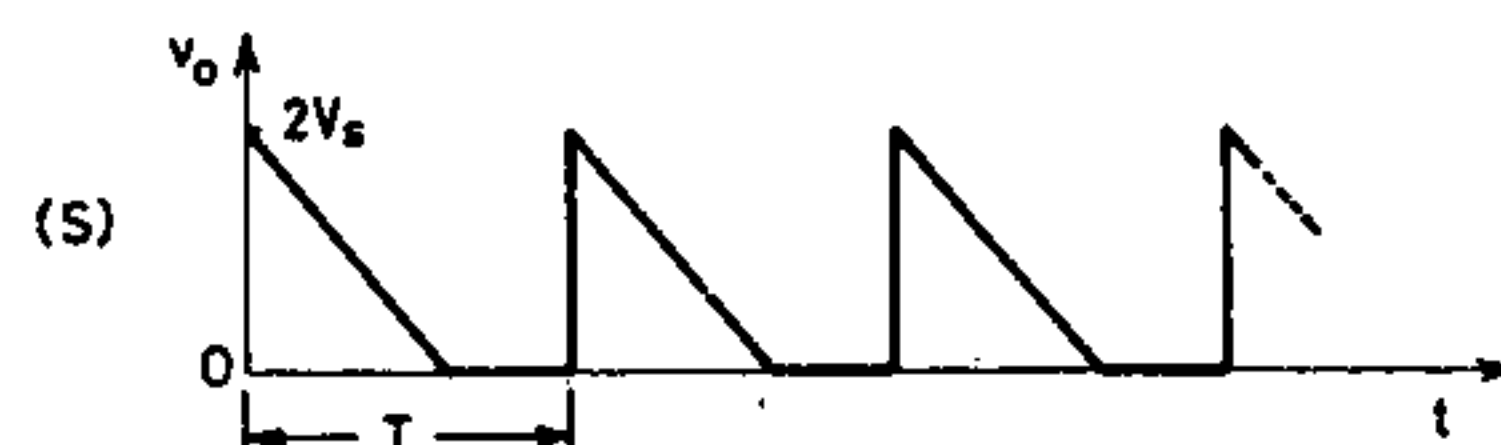
(B) Load-commutated chopper



(C) Current-commutated chopper



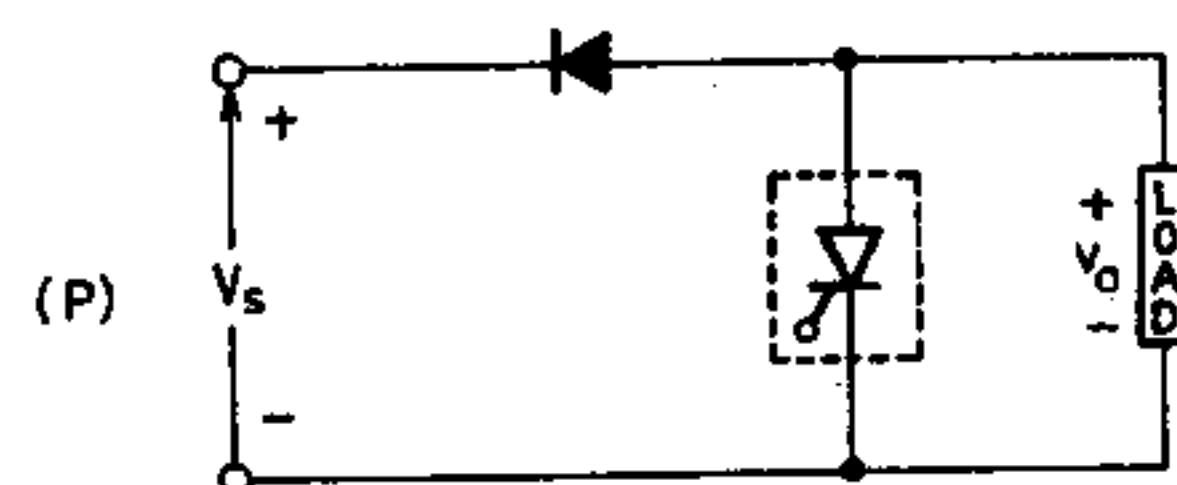
(D) Ideal dc chopper



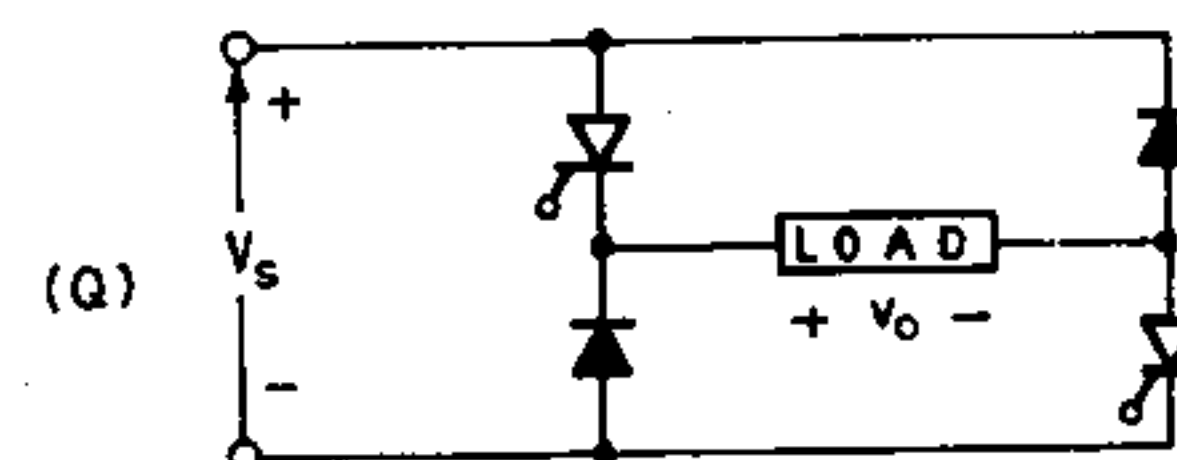
24.2. Types of choppers

Circuit Configurations

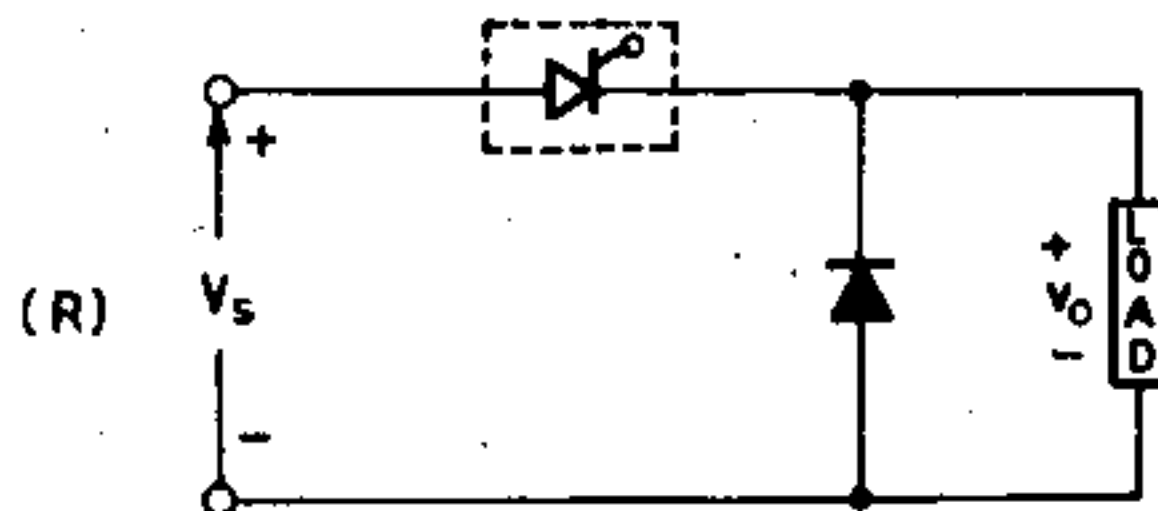
(A) Type-A chopper



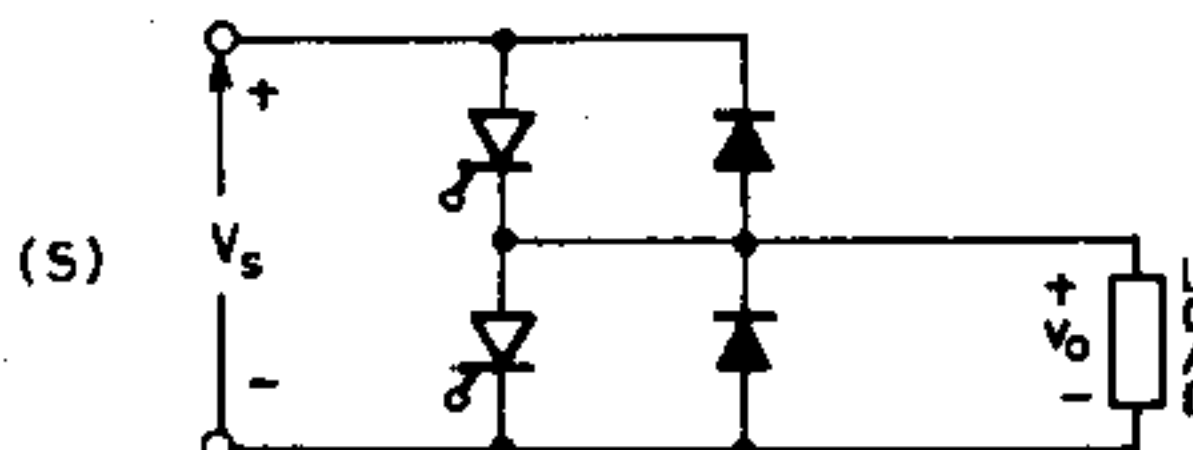
(B) Type-B chopper



(C) Type-C chopper



(D) Type-D chopper



ANSWERS

1. (d) 2. (d) 3. varied, kept constant
4. varied, kept constant, kept constant 5. (a) 6. (b) 7. (c)
8. (b) 9. (a) 10. (b) 11. (d)
12. Thyristor T1, diode D1 is reverse biased by a voltage drop of about 1.5 V across conducting thyristor T1.
13. (a) 14. 0.5 to 0.65 15. (b) 16. (b) 17. (c)
18. (c) 19. (b) 20. (a) 21. (b) 22. (d) 23.1. (b), (c)
- 23.2. (a), (b) 24.1. (A)-(Q); (B)-(S); (C)-(P); (D)-(R)
- 24.2. (A)-(R); (B)-(P); (C)-(S); (D)-(Q).

INVERTERS

1. If, for a single-phase half-bridge inverter, the amplitude of output voltage is V_s and the output power is P , then their corresponding values for a single-phase full-bridge inverter are
 - (a) V_s, P (b) $V_s/2, P/2$
 - (c) $2V_s, 2P$ (d) $2V_s, P$.
2. In voltage source inverters
 - (a) load voltage waveform v_o depends on load impedance Z , whereas load current waveform i_o does not depend on Z
 - (b) Both v_o and i_o depend on Z
 - (c) v_o does not depend on Z whereas i_o depends on Z
 - (d) both v_o and i_o do not depend upon Z .
3. A single-phase full bridge inverter can operate in load-commutation mode in case load consists of
 - (a) RL (b) RLC underdamped
 - (c) RLC overdamped (d) RLC critically damped.

4. A single-phase bridge inverter delivers power to a series connected RLC load with $R = 2 \Omega$, $\omega L = 8 \Omega$. For this inverter-load combination, load commutation is possible in case the magnitude of $1/\omega C$ in ohms is
- (a) 10 (b) 8
(c) 6 (d) zero.
5. In the half-bridge inverter of Fig. A-32, main thyristor T1 is conducting a load current. With polarity of the capacitor voltage as shown, when auxiliary thyristor TA1 is turned on, capacitor current i_c would flow through (thyristor T1/diode D1) because....

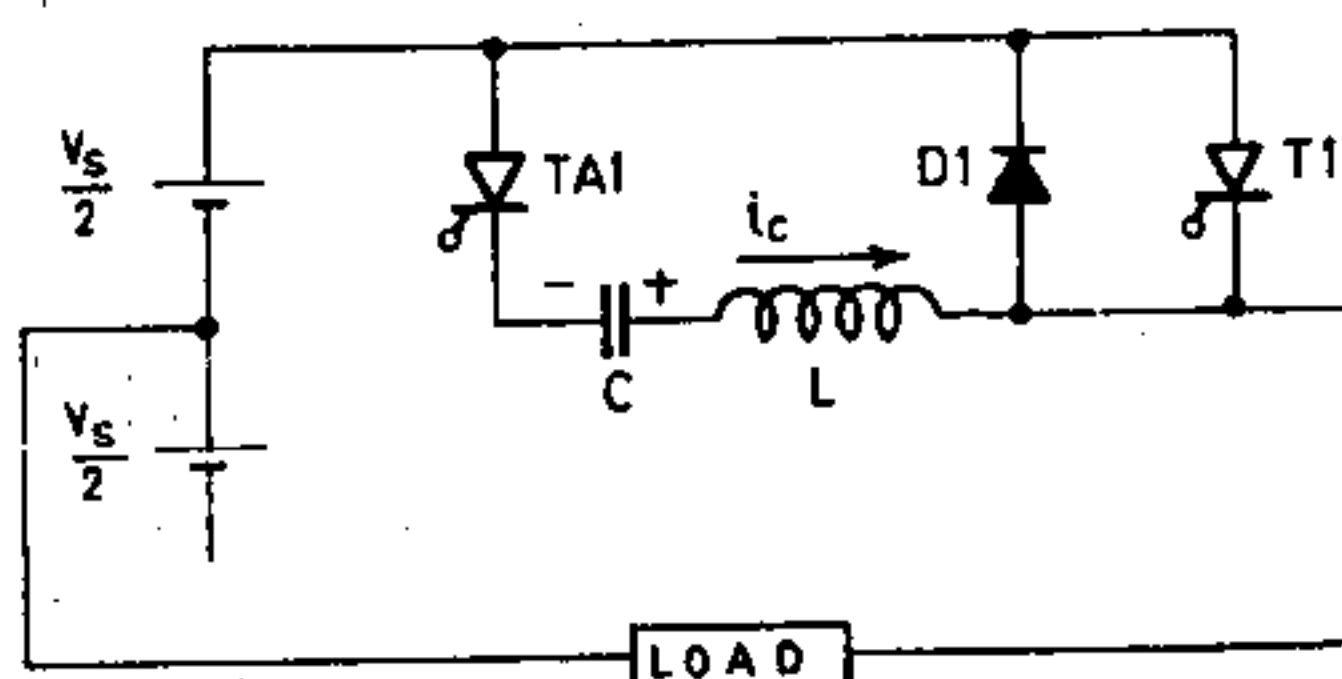


Fig. A-32

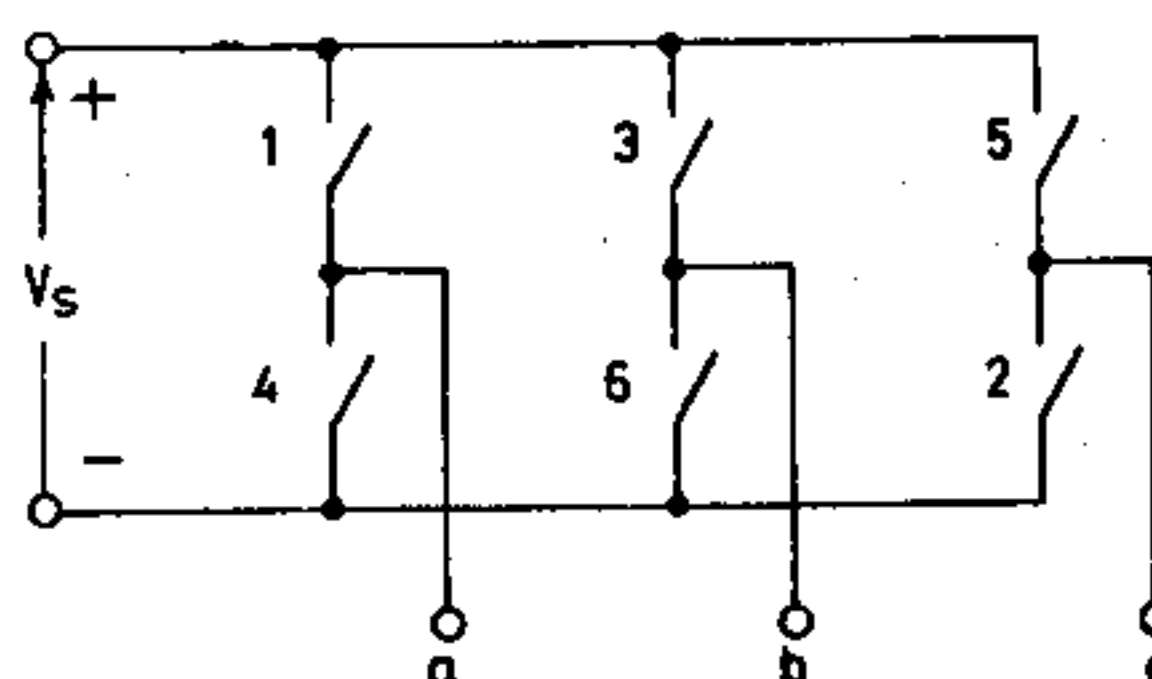


Fig. A-33

6. For a 3-phase bridge inverter in 180° conduction mode, Fig. A-33, the sequence of SCR conduction in the first two steps, beginning with the initiation of thyristor 1, is
- (a) 6, 1, 2 and 2, 3, 1 (b) 2, 3, 1 and 3, 4, 5
(c) 3, 4, 5, and 5, 6, 1 (d) 5, 6, 1 and 6, 1, 2.
7. For a 3-phase bridge inverter in 120° conduction mode, Fig. A-33, the sequence of SCR conduction in the first two steps, beginning with the initiation of thyristor 1, is
- (a) 6, 1 and 1, 2 (b) 1, 2 and 2, 3
(c) 1, 6 and 5, 6 (d) 1, 3 and 3, 4.
8. In single-pulse modulation of PWM inverters, third harmonic can be eliminated if pulse width is equal to
- (a) 30° (b) 60°
(c) 120° (d) 150°
9. In single-pulse modulation of PWM inverters, fifth harmonic can be eliminated if pulse width is equal to
- (a) 30° (b) 72°
(c) 36° (d) 108° .
10. In single-pulse modulation of PWM inverters, the pulse width is 120° . For an input voltage of 220 V dc, the r.m.s. value of output voltage is
- (a) 179.63 V (b) 254.04 V
(c) 127.02 V (d) 185.04 V.
11. In single-pulse modulation used in PWM inverters, V_s is the input dc voltage. For eliminating third harmonic, the magnitude of rms value of fundamental component of output voltage and pulse width are respectively
- (a) $\frac{2\sqrt{2}}{\pi} V_s, 120^\circ$ (b) $\frac{4 V_s}{\pi}, 60^\circ$
(c) $\frac{2\sqrt{2}}{\pi} V_s, 60^\circ$ (d) $\frac{4 V_s}{\pi}, 120^\circ$.

12. In multiple-pulse modulation used in PWM inverters, the amplitudes of reference square wave and triangular carrier wave are respectively 1 V and 2 V. For generating 5 pulses per half cycle, the pulse width should be
(a) 36° (b) 24°
(c) 18° (d) 12° .
13. In multiple-pulse modulation used in PWM inverters, the amplitude and frequency for triangular carrier and square reference signals are respectively 4 V, 6 kHz and 1 V, 1 kHz. The number of pulses per half cycle and pulse width are respectively
(a) 6, 90° (b) 3, 45°
(c) 4, 60° (d) 3, 40° .
14. In sinusoidal-pulse modulation used in PWM inverters, amplitude and frequency for triangular carrier and sinusoidal reference signals are respectively 5 V, 1 kHz and 1 V, 50 Hz. If zeros of the triangular carrier and reference sinusoid coincide, then the modulation index and order of significant harmonics are respectively
(a) 0.2, 9 and 11 (b) 0.4, 9 and 11
(c) 0.2, 17 and 19 (d) 0.2, 19 and 21.
15. Which of the following statement/statements is/are correct in connection with inverters:
(a) VSI and CSI both require feedback diodes
(b) Only CSI requires feedback diodes
(c) GTOs can be used in CSI
(d) Only VSI requires feedback diodes.
16. In a CSI, if frequency of output voltage is f Hz, then frequency of voltage input to CSI is
(a) f (b) $2f$
(c) $f/2$ (d) $3f$.
17. In sinusoidal-pulse modulation used in PWM inverters, amplitude and frequency of triangular carrier and sinusoidal reference signals are respectively 5 V, 1 kHz and 1 V, 50 Hz. If peak of the triangular carrier coincides with the zero of the reference sinusoid, then the modulation index and order of significant harmonics are
(a) 0.2, 9 and 11 (b) 0.4, 9 and 11
(c) 0.2, 17 and 19 (d) 0.2, 19 and 21.
18. In sinusoidal PWM, there are ' m ' cycles of the triangular carrier wave in the half cycle of reference sinusoidal signal. If zero of the reference sinusoid coincides with zero/peak of the triangular carrier wave, then number of pulses generated in each half cycle are respectively
(a) $(m-1)/m$ (b) $(m-1)/(m-1)$
(c) m/m (d) $m/(m-1)$.
19. In an inverter with fundamental output frequency of 50 Hz, if third harmonic is eliminated, then frequencies of other components in the output voltage wave, in Hz, would be
(a) 250, 350, 450, high frequencies
(b) 50, 250, 350, 450
(c) 50, 250, 350, 550
(d) 50, 100, 200, 250.
20. A single-phase CSI has capacitor C as the load. For a constant source current, the voltage across the capacitor is

- (a) square wave (b) triangular wave
(c) step function (d) pulsed wave.
21. A single-phase full bridge VSI has inductor L as the load. For a constant source voltage, the current through the inductor is
(a) square wave (b) triangular wave
(c) sine wave (d) pulsed wave.

ANSWERS

1. (c) 2. (c) 3. (b) 4. (a)
5. thyristor T1, diode D1 is reverse biased by the voltage drop across T1
6. (d) 7. (a) 8. (c) 9. (b) 10. (a) 11. (a)
12. (c) 13. (b) 14. (c) 15. (d) 16. (b) 17. (d)
18. (a) 19. (c) 20. (b) 21. (b)

SINGLE-PHASE VOLTAGE CONTROLLERS

- A single-phase voltage controller feeds an induction motor (A) and a heater (B)
(a) In both the loads, fundamental and harmonics are useful
(b) In A only fundamental and in B only harmonics are useful
(c) In A only fundamental and in B harmonics as well as fundamental are useful
(d) In A only harmonics and in B only fundamental are useful.
- A load resistance of $10\ \Omega$ is fed through a 1-phase voltage controller from a voltage source of $200 \sin 314 t$. For a firing angle delay of 90° , the power delivered to load in kW, is
(a) 0.5 (b) 0.75
(c) 1 (d) 2.
- A single-phase voltage controller is employed for controlling the power flow from 260 V, 50 Hz source into a load consisting of $R = 5\ \Omega$ and $\omega L = 12\ \Omega$. The value of maximum rms load current and the firing angle are respectively
(a) 20 A, 0° (b) $\frac{260}{10.91}$ A, 0°
(c) 20 A, 90° (d) $\frac{260}{10.91}$ A, 90° .
- A load, consisting of $R = 10\ \Omega$ and $\omega L = 10\ \Omega$, is being fed from 230 V, 50 Hz source through a 1-phase voltage controller. For a firing angle delay of 30° , the rms value of load current would be
(a) 23 A (b) $\frac{23}{\sqrt{2}}$ A
(c) $> \frac{23}{\sqrt{2}}$ A (d) $< \frac{23}{\sqrt{2}}$ A.
- In a single-phase voltage controller with RL load, ac output power can be controlled if
(a) firing angle $\alpha > \phi$ (load phase angle) and conduction angle $\gamma = \pi$
(b) $\alpha > \phi$ and $\gamma < \pi$
(c) $\alpha < \phi$ and $\gamma = \pi$
(d) $\alpha < \phi$ and $\gamma > \pi$.

6. A single-phase voltage controller feeds power to a resistance of $10\ \Omega$. The source voltage is 200 V rms. For a firing angle of 90° , the rms value of thyristor current in amperes is
 (a) 20 (b) 15
 (c) 10 (d) 5.
7. A single-phase voltage controller is connected to a load of resistance $10\ \Omega$ and a supply of $200 \sin 314t$ volts. For a firing angle of 90° , the average thyristor current in amperes is
 (a) 10 (b) $10/\pi$
 (c) $5\sqrt{2}/\pi$ (d) $5\sqrt{2}$.
8. A single-phase voltage controller, using two SCRs in antiparallel, is found to be operating as a controlled rectifier. This is because
 (a) load is R and pulse gating is used
 (b) load is R and high-frequency carrier gating is used
 (c) load is RL and pulse gating is used
 (d) load is RL and continuous gating is used.
9. A single-phase ac voltage controller (or regulator) fed from 50 Hz system supplies a load having resistance and inductance of $2.0\ \Omega$ and 6.36 mH respectively. The control range of firing angle for this regulator is
 (a) $0^\circ < \alpha < 180^\circ$ (b) $45^\circ < \alpha < 180^\circ$
 (c) $90^\circ < \alpha < 180^\circ$ (d) $0^\circ < \alpha < 45^\circ$.
10. For a single-phase voltage controller using two thyristors in antiparallel and with load R , draw the output voltage waveform and the voltage variation across each SCR for a firing angle of about 30° . Voltage variation across each SCR should also contain thyristor voltage drop in Fig. A-34.

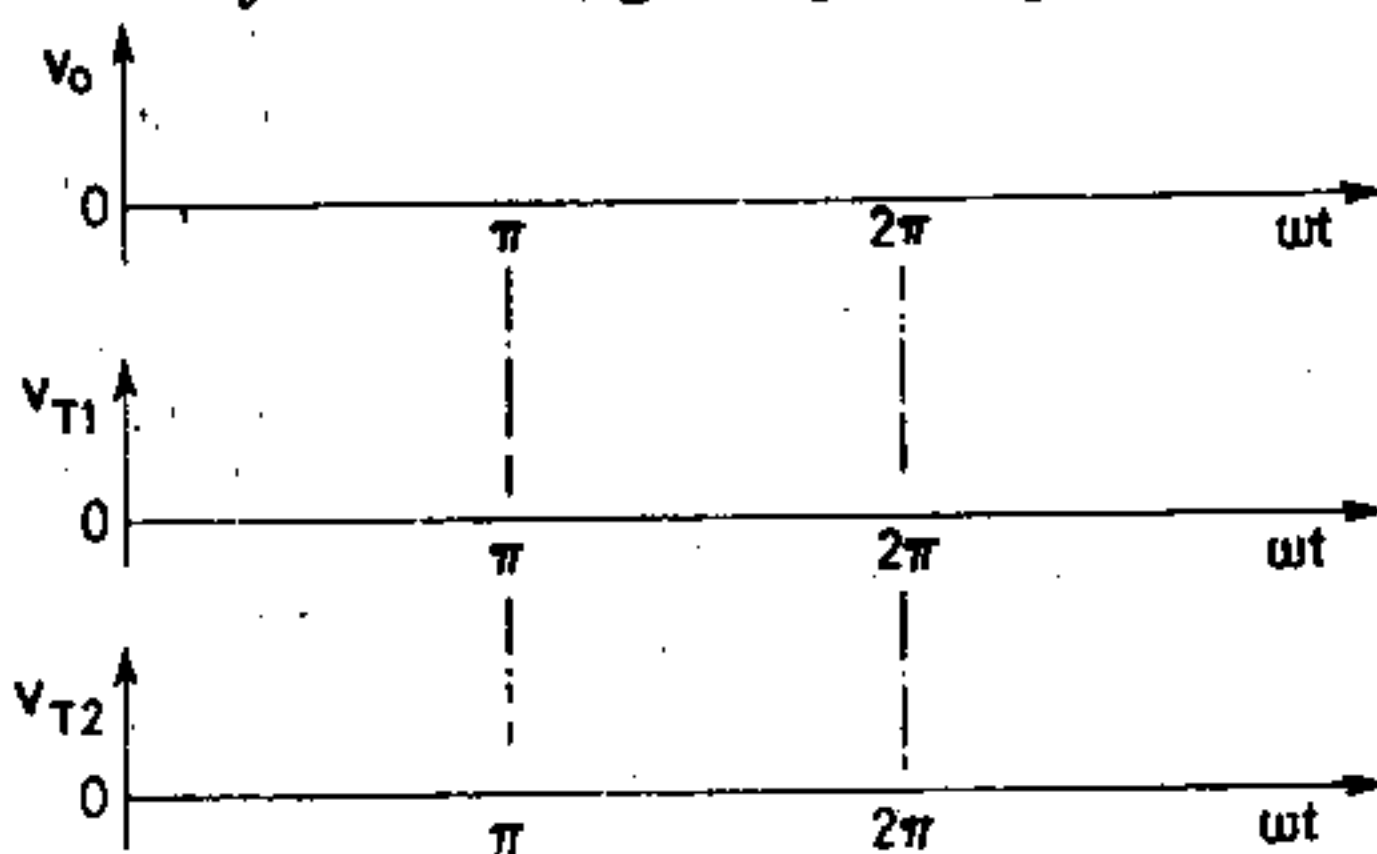


Fig. A-34

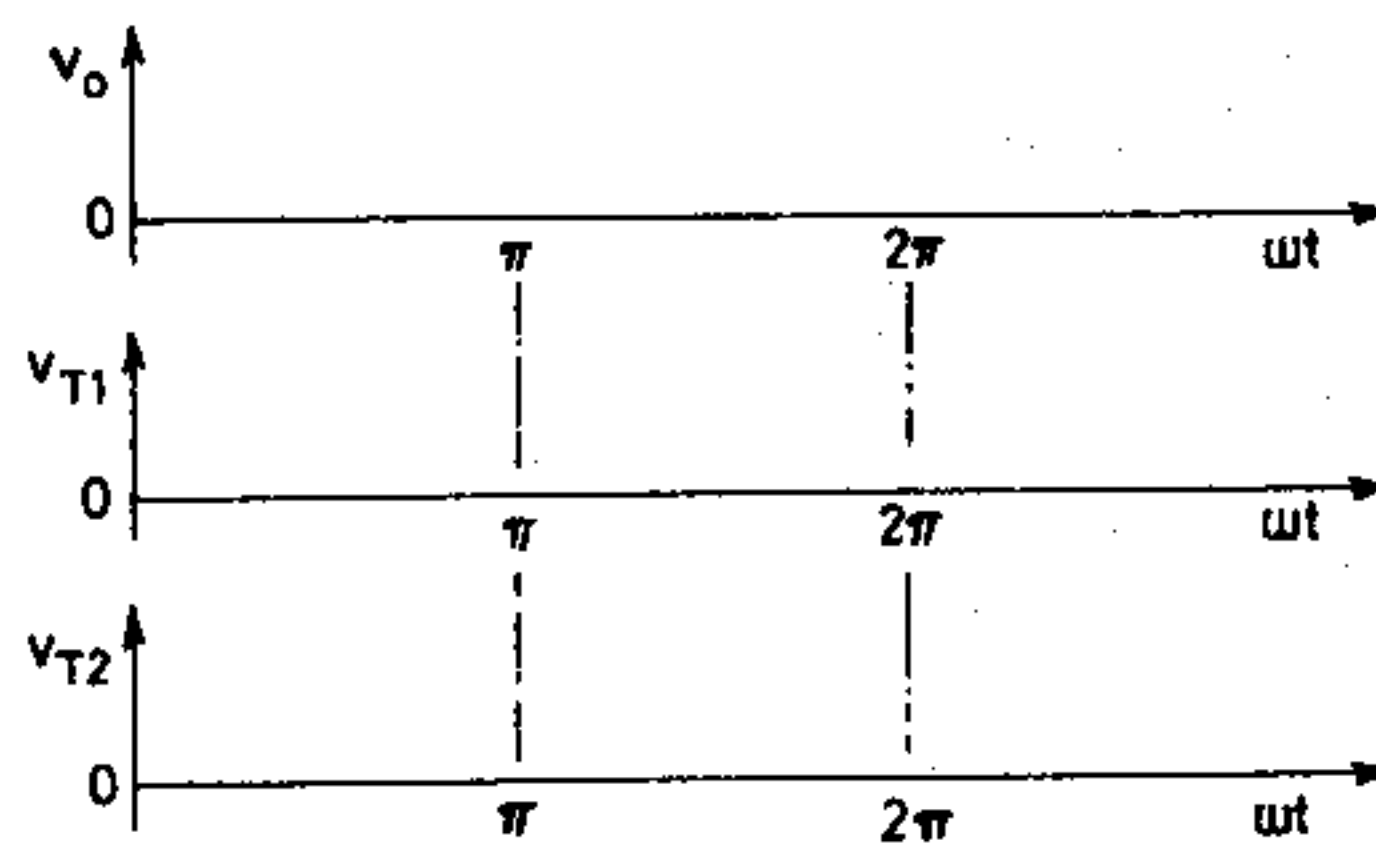


Fig. A-35

11. For a single-phase voltage controller using two thyristors in antiparallel and with RL load, draw in Fig. A-35 the output voltage waveform and the voltage variation across each SCR for a firing angle of about 30° which is greater than the load phase angle. Voltage variation across each SCR should also contain thyristor voltage drop in Fig. A-35.
12. This question has two subquestions. Amongst the four alternatives provided, one or more are correct. Write down all the correct answers unambiguously.
- 12.1. In a single-phase voltage controller with RL load, when
 (a) firing angle $\alpha < \phi$ (load phase angle), load voltage v_o is sinusoidal
 (b) $\alpha < \phi$, v_o is non-sinusoidal

(c) $\alpha > \phi$, v_o is non-sinusoidal

(d) $\alpha > \phi$, v_o is sinusoidal.

13.2. In a single-phase voltage controller with RL load, α is the firing angle, ϕ is the load phase angle and β is the extinction angle. For this voltage controller, output power can be controlled if $\alpha > \phi$ and

(a) $(\beta - \alpha) = \pi$

(b) $(\beta - \alpha) < \pi$

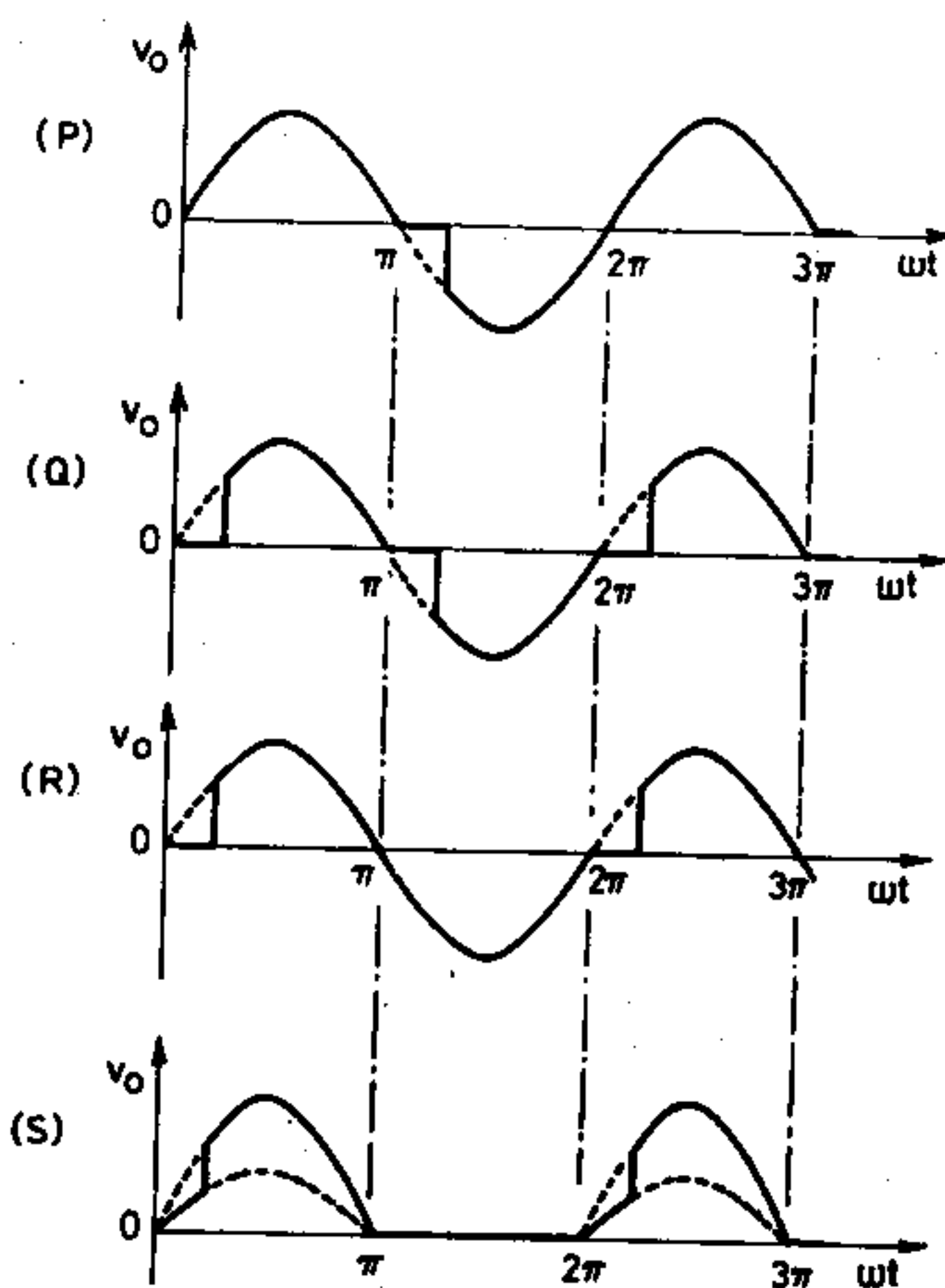
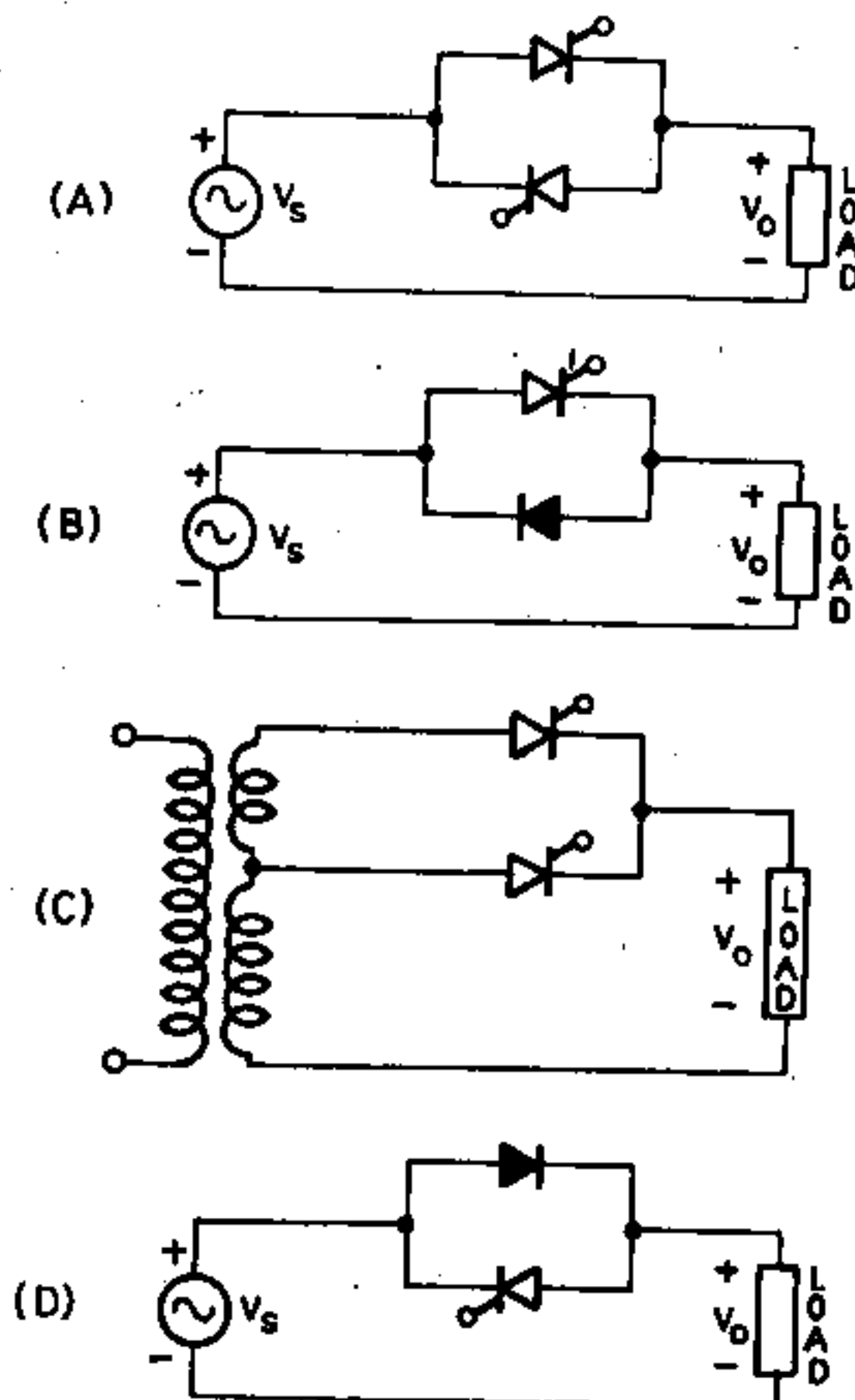
(c) $\beta > \pi$

(d) $\beta < \pi$.

13. This question contains two subquestions, each having two columns. Match every item on the left side with the most appropriate item on the right side.

13.1. Power circuit diagrams

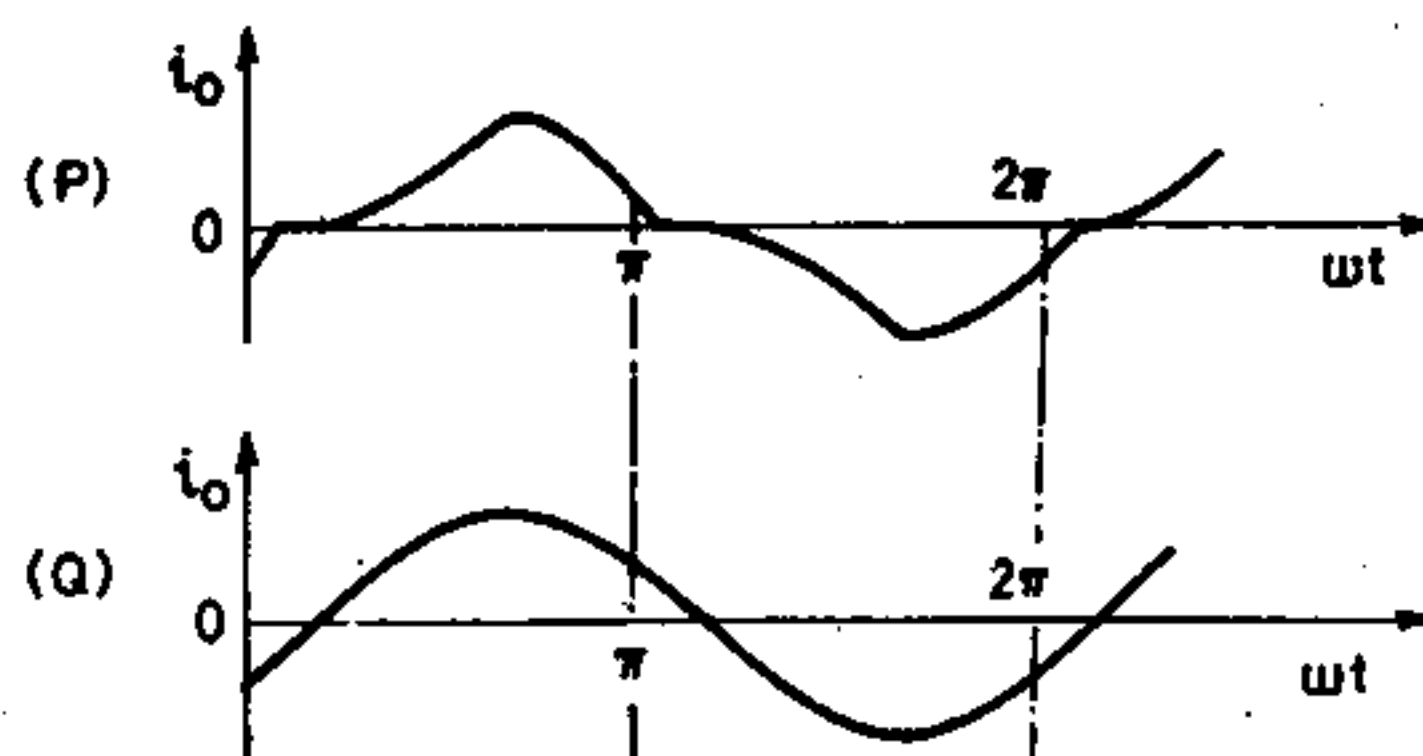
Output voltage waveforms



13.2. Firing angle = α , load phase angle = ϕ . Nature of load on single-phase voltage controllers

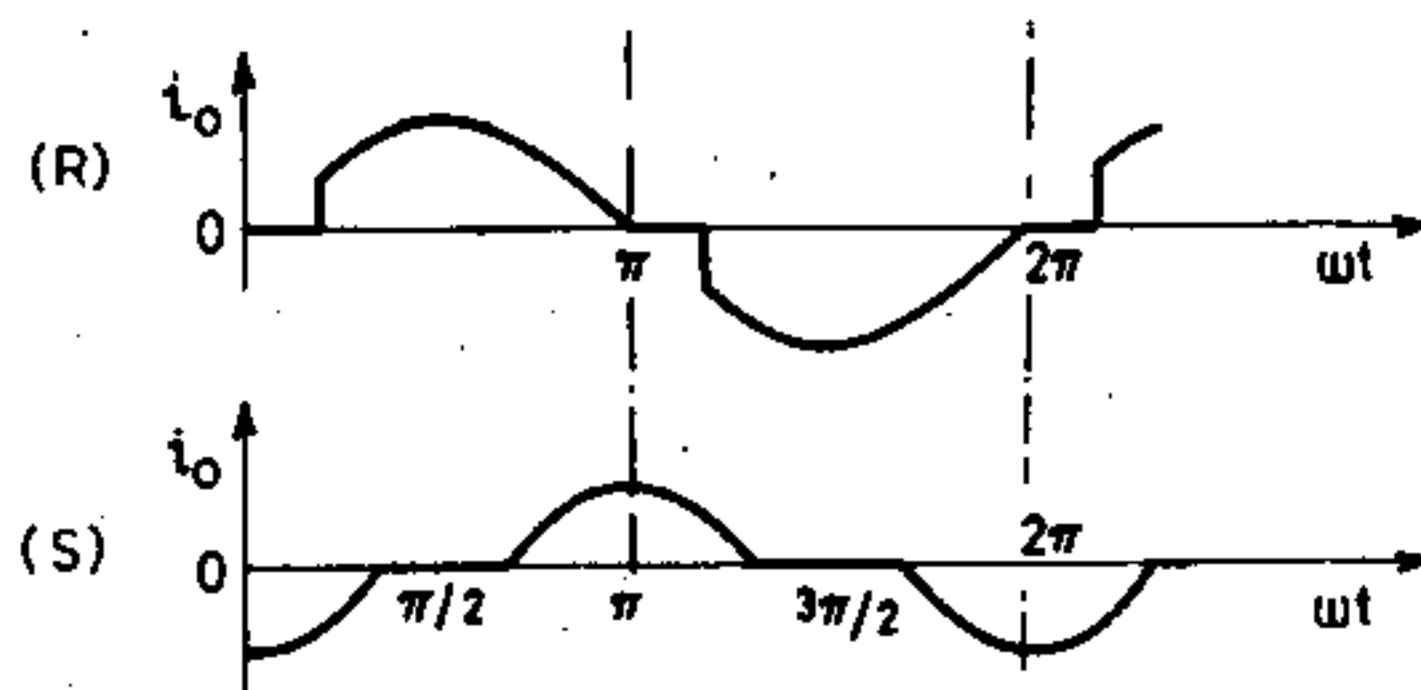
Load Current Waveforms

(A) R load



(B) RL load, $\alpha > \phi$

Load Current Waveforms

(C) RL load, $\alpha < \phi$ (D) L load, $\alpha > \phi$ 

ANSWERS

- | | | | | | |
|--|--------|----------------|---|--------|--------|
| 1. (c) | 2. (c) | 3. (a) | 4. (b) | 5. (b) | 6. (c) |
| 7. (b) | 8. (c) | 9. (b) | 10. See the book | | |
| 11. See the book | | 12.1. (a), (c) | 12.2. (b), (c) | | |
| 13.1. (A)-(Q); (B)-(R); (C)-(S); (D)-(P) | | | 13.2. (A)-(R); (B)-(P); (C)-(Q); (D)-(S). | | |

CYCLOCONVERTERS

- A cycloconverter is a
 - frequency changer (f_c) from higher to lower frequency with one-state conversion
 - f_c from higher to lower frequency with two-stage conversion
 - f_c from lower to high frequency with one-state conversion
 - either (a) or (c).
- The cycloconverters (CCs) require natural or forced commutation as under :
 - natural commutation in both step-up and step-down CCs
 - forced commutation in both step-up and step-down CCs
 - forced commutation in step-up CCs
 - forced commutation in step-down CCs.
- For converting 3-phase supply at one frequency to single-phase supply at a lower frequency, the basic principle is to....(vary/keep) the firing angle..... (constant/gradually).
- Three-phase to three-phase cycloconverters employing 18 SCRs and 36 SCRs have the same voltage and current ratings for their component thyristors. The ratio of VA rating of 36-SCR device to that of 18-SCR device is
 - $\frac{1}{2}$
 - 1
 - 2
 - 4.
- Three-phase to 3-phase cycloconverters employing 18 SCRs and 36 SCRs have the same voltage and current ratings for their component thyristors. The ratio of power handled by 36-SCR device to that handled by 18-SCR device is

(a) 4	(b) 2
(c) 1	(d) $\frac{1}{2}$.

6. The number of thyristors required for single-phase to single-phase cycloconverter of the mid-point type and for three phase to three-phase 3-pulse type cycloconverter are respectively

- (a) 4, 6
- (b) 8, 18
- (c) 4, 18
- (d) 4, 36.

7. A 3-phase to single-phase conversion device employs a 6-pulse bridge cycloconverter. For an input voltage of 200 V per phase, the fundamental rms value of output voltage is

- (a) $600/\pi$ V
- (b) $300\sqrt{3}/\pi$ V
- (c) $300/\pi$ V
- (d) $600\sqrt{3}/\pi$ V.

8. The question has three subquestions. Amongst the four alternatives provided, one or more are correct. Write down all the correct answers unambiguously.

8.1. A three-phase to single-phase cycloconverter consists of positive and negative group of converters. In this device one of the two component converters would operate as a

- (a) rectifier if the output voltage V_0 and output current I_0 have the same polarity
- (b) inverter if V_0 and I_0 have the same polarity
- (c) rectifier if V_0 and I_0 are of opposite polarity
- (d) inverter if V_0 and I_0 are of opposite polarity.

8.2. A 3-phase to 3-phase cycloconverter requires

- (a) 18 SCRs for 3-phase device
- (b) 18 SCRs for 6-pulse device
- (c) 36 SCRs for 3-pulse device
- (d) 36 SCRs for 6-pulse device.

8.3. Which of the following statements are correct for cycloconverters ?

- (a) Step-down cycloconverter (cc) works on natural commutation
- (b) Step-up cc requires forced commutation
- (c) Load commutated cc works on line commutation
- (d) Load commutated cc requires a generated emf in the load circuit.

9. This question contains four subquestions, each having two columns. Match every item on the left side with the most appropriate item on the right side.

9.1. Power electronic controller

- (A) Controlled rectifier
- (B) Chopper
- (C) Cycloconverter
- (D) Inverter

Applications

- (P) Aircraft supplies
- (Q) Electric car
- (R) Induction heating
- (S) Rolling mill drive.

9.2. Power electronic controller

- (A) Controlled rectifier
- (B) Voltage controller
- (C) Cycloconverter
- (D) Inverter

Applications

- (P) High-power ac drive
- (Q) Solar cells
- (R) Ceiling fan drive
- (S) Magnet power supply.

9.3. Power electronic controller

- (A) Inverter
- (B) Controlled rectifier
- (C) Voltage controller
- (D) Chopper

Applications

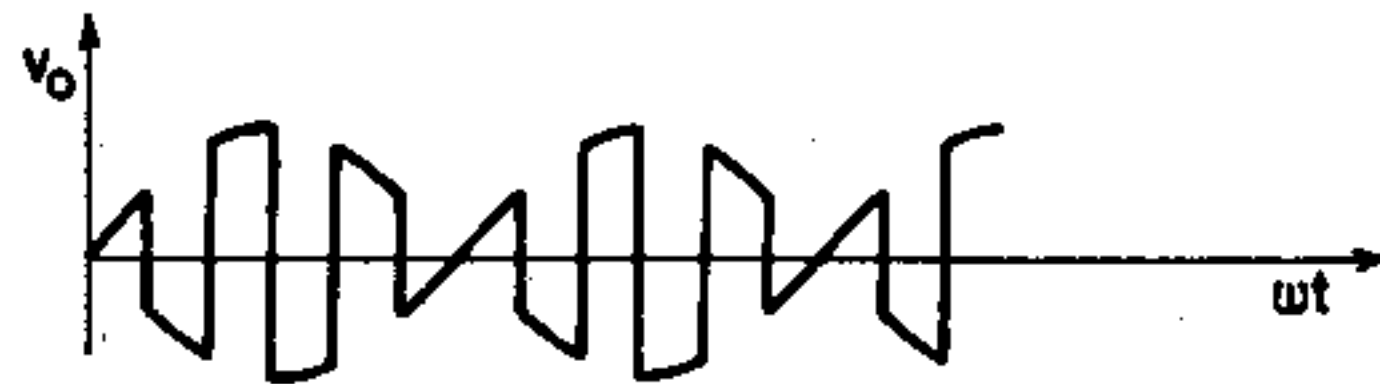
- (P) Fork-lift truck
- (Q) Illumination control
- (R) Uninterruptible power supply
- (S) Hydrogen production.

9.4. Types of cycloconverters

Output Voltage Waveforms

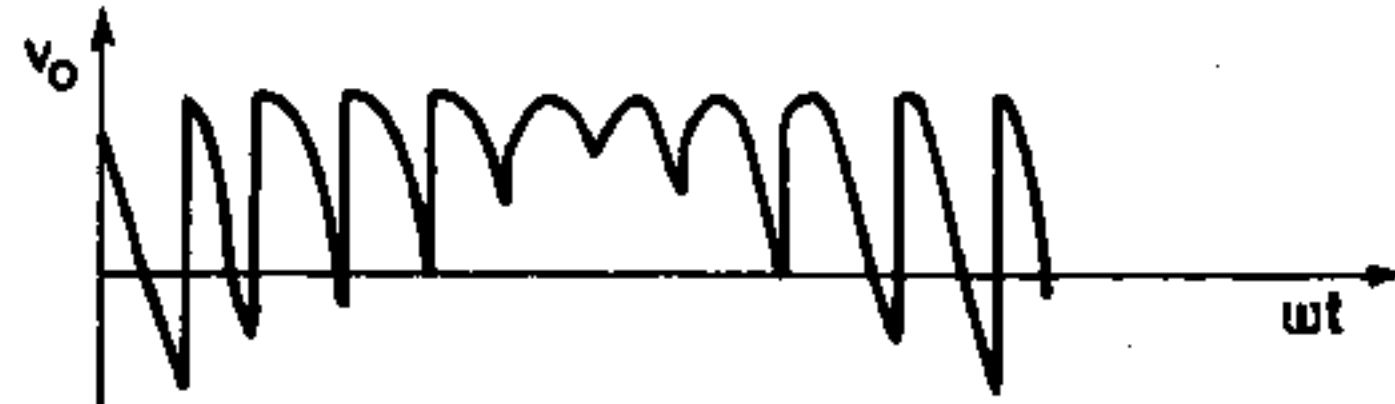
(A) 1-phase to 1-phase with continuous conduction

(P)



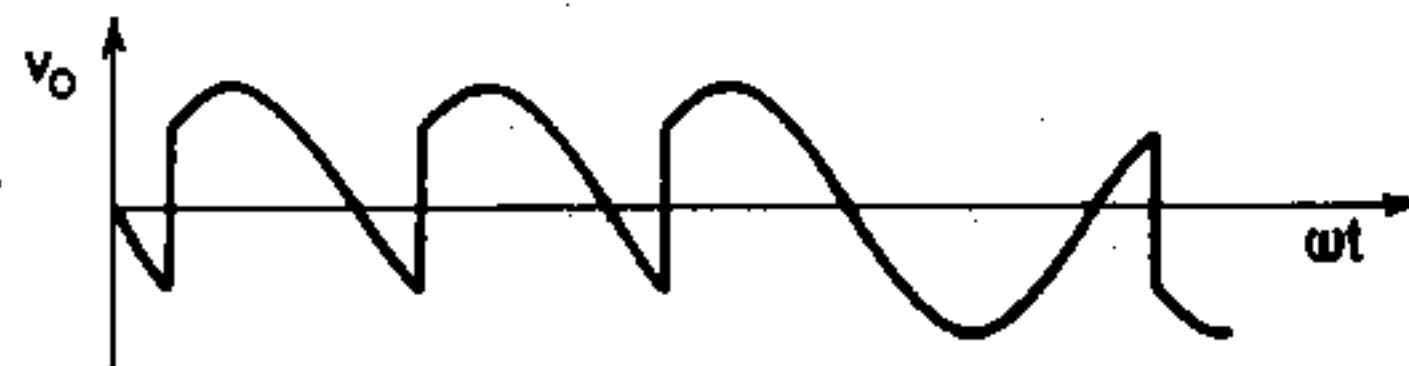
(B) 1-phase to 1-phase with discontinuous conduction

(Q)



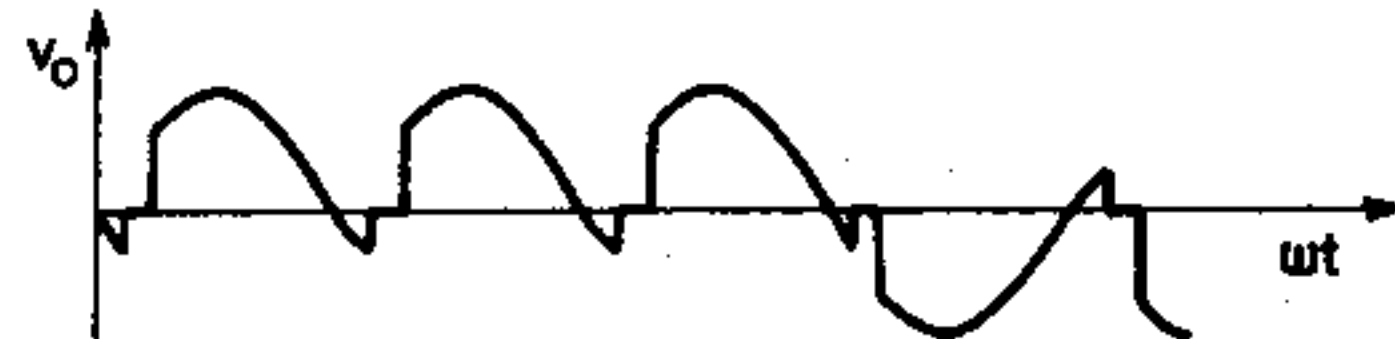
(C) Step-up device

(R)



(D) 3-phase to 1-phase device

(S)



ANSWERS

- | | | | | |
|---|--------|--------------------|---|--------------------|
| 1. (d) | 2. (c) | 3. vary, gradually | 4. (c) | 5. (a) |
| 6. (c) | 7. (a) | 8.1. (a), (d) | 8.2. (a), (d) | 8.3. (a), (b), (d) |
| 9.1. (A)-(S); (B)-(Q); (C)-(P); (D)-(R) | | | 9.2. (A)-(S); (B)-(R); (C)-(P); (D)-(Q) | |
| 9.3. (A)-(R); (B)-(S); (C)-(Q); (D)-(P) | | | 9.4. (A)-(R); (B)-(S); (C)-(P); (D)-(Q) | |

REFERENCES

1. SCR Manual, 5th Edition, N.Y., General Electric Company, 1972.
2. F. Csaki et. al., 'Power Electronics', Budapest : Akademiai Kiado, 1975.
3. F.E. Gentry et. al., 'Semiconductor Controlled Rectifiers', Prentice-Hall of India New Delhi, 1964.
4. B.K. Bose, 'Power Electronics and AC Drives', Prentice-Hall, Englewood Cliffs, New Jersey 07632, 1986.
5. P.C. Sen, 'Thyristorised DC Drive', New York : Wiley Interscience, 1981.
6. J.M.D. Murphy and F.G. Turnbull 'Power Electronic Control of AC Motors', Pergamon Press, Oxford, 1988.
7. B.K. Bose, 'Evaluation of Modern Power Semiconductor Devices and Future Trends of Converters', IEEE Trans. Industry Applications, vol. 28, No. 2, pp. 403 – 413, March/April, 1992.
8. NED MOHAN et. al. , 'Power Electronics', John Wiley and Sons, 1989.
9. M.H. Rashid, 'Power Electronics', Prentice-Hall of India, New Delhi, 1993.
10. M. Chilikan, 'Electric Drive', Mir Publishers, Moscow, 1970.